

Australian Household Consumption, Wealth and Credit Conditions: An Updated Single-Equation LIVES Estimate

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1 Australian Household Consumption, Wealth and Credit Conditions: An Updated Single-Equation LIVES Estimate

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1.1 Abstract

We estimate a single-equation Muellbauer–Williams LIVES consumption model for Australia over 1988Q4–2024Q4, extending Williams (2010) and Muellbauer and Williams (2012) by sixteen years of post-GFC data and applying contemporary identification methods. Wealth is disaggregated into housing, illiquid financial (equities plus superannuation), and net liquid assets (deposits net of total household debt), and we adopt the Jordà (2005) local-projection permanent-income forecaster used in the Italian implementation of De Bonis, Marino and Muellbauer (2024). The cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ is accepted in every disaggregated specification and sample window we estimate, validating the Italian convention.

The preferred specification delivers a speed of adjustment $\lambda = -0.180$ (vs Williams' published -0.286) and structural long-run coefficients on individual wealth components that are broadly consistent with Williams' Table 1 estimates: $\gamma_{HA} = 0.049$ (Williams: 0.049), $\gamma_{IFA} = 0.030$ (Williams: 0.022), $\gamma_{NLA} = 0.196$ (Williams: 0.159). The OLS coefficients are 14–37 per cent below Williams' implied OLS values, but our smaller $|\lambda|$ scales them up to a structural γ profile in line with the FIML estimates. The often-noted "Australian permanent-income puzzle" — a significantly negative long-run coefficient on $\log(y^p/y)$ under a rolling AR(8) forecaster — flips to a positive $+0.30$ under Italy LP, matching Williams' calibrated $+0.20$ in sign and broad magnitude. Adding the time-varying housing-wealth m.p.c. interaction to the Williams CCI interactions specification yields $\lambda = -0.377$, exceeding Williams' calibrated value in magnitude.

We assemble a back-extended master dataset to 1976Q3 — using a TRYM long-run house-price series, RBA D03 monetary aggregates, RBA D02 total credit, historical labour-force compilations, and documented aggregate and disaggregated wealth proxies anchored at 1988Q3 — to test whether sample length is the binding constraint on tighter agreement with Williams. Refitting the disaggregated no-CCI specification on the back-extended 1976Q3–2024Q4 sample ($n = 190$), λ moves 37 per cent closer to Williams ($-0.140 \rightarrow -0.193$) but individual wealth coefficients become smaller rather than larger, and the net-liquid coefficient collapses toward zero. The CCI placebo battery fails on both samples: the literal Williams 4-knot deteriorates from the 34th adjusted- R^2 percentile on the 1988+ sample to the 19th on the back-extended sample, the maximal-GETS reduction sits at the 64th percentile, and

the sectional-period specification sits at the 36th. A two-equation SUR of consumption and house prices on the back-extended sample finds residual correlation $\hat{\rho} \approx 0.0007$, so joint estimation delivers no efficiency gain at the quarterly frequency. A three-equation joint cross-equation CCI identification retains only two of six single-equation knot survivors. We read these findings as indicating that the structural identification Williams (2010) delivers comes from cross-equation parameter restrictions in his four-equation FIML system rather than from sample length, knot count, or sign-prior structure; the headline single-equation estimate nevertheless reproduces his structural coefficient profile to a useful approximation.

The paper includes a structured robustness suite covering instrumental variables, joint SUR estimation, Chow tests at multiple break dates, the Drehmann (2017) amortising-mortgage adjustment, the Williams smoothed-step credit-conditions spline with maximal-GETS, Kalman state-space, and sectional sign-prior alternatives, rolling-window estimation, out-of-sample forecast validation against random-walk and AR(1) benchmarks, and back-extension robustness on Spec 1 and Spec 4. A full reproducibility kit accompanies the paper, along with a multi-equation LIVES scaffold that lays foundations for a companion paper.

1.2 1. Introduction

1.2.1 1.1 Motivation

Australian household consumption faces a set of policy and analytical questions that the standard New Keynesian DSGE workhorse — with its representative consumer, exogenous wealth process, and assumed-frictionless credit access — is structurally ill-equipped to answer. Among them: how much of the post-2008 moderation in consumption growth is attributable to the macroprudential tightening of 2014 and 2017, and how much to the post-GFC household balance-sheet repair? How sensitive is consumption to housing wealth at different points in the credit cycle, when households are or are not able to extract home equity? Did the COVID shock and JobKeeper income support move permanent-income expectations, or only the short-run quarter? And how should a central bank think about the wealth channel of monetary policy when much of household wealth is housing wealth, mortgage debt is at near-historic levels relative to income, and the credit conditions in which households operate have evolved markedly since financial deregulation in the 1980s?

The Muellbauer-Williams "LIVES" framework — Latent Interactive Variable Equation System, after Tobin's flow-of-funds tradition (Duca and Muellbauer, 2013) — was developed to answer exactly these questions. It augments the standard credit-augmented life-cycle consumption function (Friedman 1957, Ando-Modigliani 1963, Tobin-Dolde 1971) with: (i) wealth disaggregated into liquid, illiquid financial and housing components, recognising that they have different marginal propensities to consume (Backus and Purvis 1980); (ii) a latent credit-conditions index (CCI) that interacts with key parameters — the housing-wealth m.p.c. that is amplified by easier credit, the down-payment constraint that is relaxed by easier credit, the role of permanent-income expectations that rises with credit ease — and (iii) a multi-equation system that estimates this latent factor jointly across consumption, house prices, mortgage stock, and home equity withdrawal under cross-equation parameter restrictions.

Williams (2010) applied this framework to Australia for 1978–2008, producing the canonical Australian LIVES estimate. Sixteen years of post-GFC and post-COVID data have since

accumulated, and a contemporary central-bank-quality update of his framework is warranted. This paper provides that update.

1.2.2 1.2 Contribution

This paper makes four contributions to the Australian household consumption literature.

An updated Williams (2010) LIVES estimate to 2024Q4 using publicly available data, with the permanent-income forecaster shifted from a rolling AR(8) — the implementation in earlier Australian work — to the Jordà (2005) local projection with a labour-force-share predictor, following the Italian implementation of De Bonis, Marino and Muellbauer (2024). Under Italy LP the speed of adjustment is within 40 per cent of Williams' published value, and the structural γ profile on the disaggregated wealth components is broadly consistent with Williams' Table 1. The often-noted "Australian permanent-income puzzle" — a significantly negative long-run coefficient on $\log(y^p/y)$ under the AR forecaster — flips to positive under Italy LP, matching Williams' calibrated value in sign and broad magnitude.

A back-extended master dataset (1976Q3–2024Q4, $n = 194$ quarters) with documented growth-rate splices for house prices (Treasury TRYM historical compilation, 1959Q3+), the M3 monetary aggregate (RBA D03, 1959Q3+), total credit (RBA D02, 1976Q3+), and labour force (historical compilation, 1964Q3+). For the 1976Q3–1988Q2 window where ABS sectoral household balance-sheet data are unavailable, we construct aggregate and disaggregated wealth proxies anchored at 1988Q3 (§3.9–§3.13).

A direct test of whether the residual coefficient gap with Williams is a sample-length artefact. Refitting the disaggregated no-CCI specification (Spec 4) on the back-extended sample, λ moves 37 per cent closer to Williams ($-0.140 \rightarrow -0.193$) but individual wealth coefficients become smaller rather than larger; γ_{NLA} collapses to near zero and γ_{EQ} retains a wrong sign. Triangulating with random-knot placebo tests on both samples, with a Zellner SUR of consumption and house prices showing essentially zero residual correlation, and with a three-equation joint cross-equation CCI identification that retains only two of six single-equation knot survivors, we read the binding constraint on tighter agreement with Williams as the single-equation framing itself rather than the post-1988 sample window. The multi-equation framework is laid out in a separate companion directory.

A structured robustness suite mirroring the De Bonis et al. (2024) Italian methodology: instrumental variables, joint SUR, multi-window Chow tests, the Drehmann (2017) amortisation adjustment, scaled-income and Williams non-property-income robustness, an AR/Italy LP comparison column, a permanent-income filter sensitivity grid, COVID-period stability tests, rolling-window estimation, out-of-sample forecast validation against random-walk and AR(1) benchmarks, and back-extension robustness columns for Spec 1 and Spec 4. The credit-conditions spline is stress-tested against random-knot placebos under three identification protocols (literal Williams 4-knot, maximal-GETS reduction, sectional sign priors) on both samples. The empirical pattern is that none of these specifications strongly outperforms random-knot placements on the post-deregulation Australian data.

1.2.3 1.3 Headline result

Under the Italy local-projection PI forecaster the preferred Spec 6 delivers a speed of adjustment $\lambda = -0.180$ on a non-overlapping sample (1988Q4–2024Q4 vs Williams' 1978Q1–2008Q2), about 63 per cent of Williams' published -0.286 . The structural long-run coefficients

on the disaggregated wealth components — $\gamma_{HA} = 0.049$, $\gamma_{IFA} = 0.030$, $\gamma_{NLA} = 0.196$ — are within ± 37 per cent of Williams' Table 1 values in every case, with γ_{HA} matching Williams almost exactly. Adding the time-varying housing-wealth m.p.c. interaction in the Williams CCI-interactions specification (Spec 8) yields $\lambda = -0.377$, *exceeding* Williams' calibrated value in magnitude. The cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ is accepted in every disaggregated specification and sample window we estimate, validating the Italian convention. The "Australian permanent-income puzzle" is resolved by the methodology shift from AR to Italy LP: the long-run coefficient on $\log(y^p/y)$ flips from -0.20 to $+0.30$, matching Williams' calibrated $+0.20$ in sign and broad magnitude.

The CCI placebo evidence is sharper. The literal Williams 4-knot specification sits at the 34th adjusted- R^2 percentile on the 1988+ sample and deteriorates to the 19th percentile on the back-extended 1976Q3+ sample. The maximal-GETS canonical sits at the 64th/36th percentile on the extended sample. The sectional sign-prior specification — Williams' Aust paper §5.1 implementation — sits at the 36th/40th, no stronger than random period placements. The three-equation joint cross-equation identification explains the single-equation placebo weakness: when knots are required to satisfy sign priors *jointly* across consumption, house prices and mortgage stock, only two of six single-equation survivors pass — 1986 financial deregulation and 2017 APRA macroprudential round II. The consumption-fitted CCI is therefore not a true common factor; four of its six surviving knots are consumption-equation specific. The path to closing the residual gap with Williams' system estimates runs through a full FIML build with cross-equation parameter restrictions, which the single-equation framework adopted here cannot deliver.

1.2.4 1.4 Roadmap

Section 2 surveys the LIVES literature. Section 3 documents data construction, including the back-extension sources and proxies (§§3.9–3.13). Section 4 presents the model. Section 5 develops identification of credit conditions, including the placebo battery and the cross-equation joint-identification test. Section 6 sets out the eleven specifications and the four-screen selection rubric. Section 7 presents preferred-specification results and compares the implied structural coefficient profile with Williams' Table 1. Section 8 runs the structured robustness suite, including the back-extension robustness columns and the multi-equation SUR findings. Section 9 reconciles our estimates with Williams' published values and discusses the residual gap. Section 10 presents the long-run decomposition and policy implications. Section 11 concludes.

1.3 2. Literature review

This paper sits at the intersection of three literatures: the Muellbauer-Williams "LIVES" tradition that integrates wealth, credit conditions and life-cycle behaviour into consumption equations; the Australian empirical consumption literature, which has historically focused on aggregate wealth effects without the explicit credit-conditions machinery; and the small but growing body of work that disciplines permanent-income measurement using forecasting approaches more robust than the standard $AR(p)$ recipe. We review each in turn before placing the present contribution.

1.3.1 2.1 Theoretical foundations

The empirical specification estimated below has its origins in the Davidson-Hendry-Srba-Yeo (1978) error-correction consumption function and the canonical permanent-income hypothesis (PIH) of Friedman (1957) and Hall (1978). The DHSY approach embeds a long-run cointegrating relationship between consumption, income and wealth into a short-run dynamic equation in growth rates, with an error-correction term governing the speed at which the consumption ratio reverts to its long-run determinants. Engle and Granger (1987) provide the formal econometric framework, with Hendry and Krolzig (2005) and Doornik (2009) refining the general-to-specific reduction methodology that disciplines the choice of short-run regressors.

A long line of empirical work has documented systematic departures from strict PIH behaviour. Campbell and Mankiw (1989, 1991) show that roughly half of US consumption tracks current rather than permanent income, consistent with rule-of-thumb behaviour or liquidity constraints. Carroll and Kimball (1996) establish the concavity of the consumption function under prudence, providing a microeconomic basis for differential marginal propensities to consume across wealth classes. Carroll (2001) and Deaton (1992) develop the buffer-stock interpretation that motivates the distinction between liquid and illiquid wealth in the Muellbauer specifications used here. The empirical implication — that wealth components with different liquidity, transactions costs, and ownership concentration should enter the consumption equation with different coefficients — is the cornerstone of the disaggregated specifications (Specs 4–7) reported in §7.

1.3.2 2.2 The LIVES framework

Muellbauer (2007), originally prepared for the Federal Reserve Bank of Kansas City Jackson Hole symposium, sets out the integrated approach to modelling consumption with wealth, housing collateral and credit conditions. The paper's central insight is that credit-market liberalisation interacts with the wealth and interest-rate channels: when credit is tight, housing wealth collateralises borrowing only weakly and the down-payment hurdle dampens consumption; when credit is loose, housing wealth and current consumption become more tightly linked. This insight requires a credit-conditions index (CCI) entering both as a long-run intercept shifter and as a multiplicative interaction with the wealth, interest-rate and permanent-income terms.

The framework was operationalised in a series of country studies. Aron, Duca, Muellbauer, Murata and Murphy (2012) jointly estimate the framework on Japan, the United Kingdom and the United States, finding consistent positive long-run housing wealth effects in the UK and US (where home equity withdrawal is institutionally available) and a much smaller effect in Japan (where it is not). Duca, Muellbauer and Murphy (2010) apply the framework to the global financial crisis, demonstrating that the abrupt tightening in CCI from 2007 quantitatively rationalises the sharp consumption pullback observed in heavy-MEW economies. Duca, Muellbauer and Tobin (2013), in *European Economy* Discussion Paper 14, formalise what they label the "LIVES" approach — the *life-cycle Integration of Variable Expectations and Structure* — emphasising the joint determination of consumption, house prices, mortgage debt and home equity withdrawal in a four-equation system identified by common factors and cross-equation sign restrictions.

Two parallel implementations frame the present paper. De Bonis, Marino and Muellbauer (2024) estimate a single-equation Italian adaptation that imposes the cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ (deposits and household debt enter with equal-and-opposite coefficients), adopts a Jordà (2005) local-projection permanent-income forecaster, applies a Drehmann (2017) amortising-mortgage adjustment to the real mortgage rate, and validates the single-equation OLS estimator against joint SUR. Chauvin and Muellbauer (2018), in the Banque de France working-paper series with an associated online complement, undertake a similar France adaptation, with particular attention to the institutional differences — limited home equity withdrawal, a larger social housing sector — that shape the housing-wealth channel. Both papers explicitly take Williams' Australian work as a methodological precedent; the present paper closes the loop by applying the Italian methodology back to the Australian data on which the original LIVES Australia estimation was performed.

1.3.3 2.3 The Australian application and prior Australian consumption work

The Australian application was developed in two companion papers by David Williams during his Oxford doctorate. Williams (2009), Oxford Economics Series Working Paper 432, focuses on the house-price equation and its identification under financial liberalisation; the paper develops the spline-based credit-conditions index from STAMP (Koopman et al. 2000) unobserved-components estimation, anchors the index at the four institutional turning points 1979/1992/1998/2007, and constructs the non-property household disposable income measure that we replicate in §3.6. Williams (2010), Oxford Economics Series Working Paper 492, estimates the four-equation LIVES system (consumption, house prices, mortgage stock, home equity withdrawal) on Australian data 1977Q2–2008Q2. The published version of this work, Muellbauer and Williams (2012) "Credit conditions and the real economy: the elephant in the room", appears as the lead chapter of *BIS Papers* No. 64 and constitutes our primary benchmark.

Australian consumption modelling outside the LIVES tradition has a long history. Tan and Voss (2000), in RBA Research Discussion Paper 2000-09, estimate aggregate-wealth effects on Australian consumption using ABS National Accounts and RBA balance-sheet data, finding significant positive effects of both housing and financial wealth. Dvornak and Kohler (2003), in RBA RDP 2003-07, use a state-level panel to identify wealth effects from cross-state variation, finding larger marginal propensities to consume out of stock-market wealth than out of housing wealth, in apparent contrast to the time-series evidence; their findings are partly reconciled by the Muellbauer–Williams framework once credit-conditions interactions are introduced.

The Reserve Bank of Australia's macroeconometric model MARTIN, introduced in Cusbert and Kendall (2018) in the RBA Bulletin and documented in Ballantyne et al. (2019) in RBA RDP 2019-07, includes a household consumption block that incorporates wealth effects and credit conditions in a more reduced-form way than the LIVES specification. The MARTIN consumption equation imposes calibrated elasticities for several channels rather than estimating the full long-run cointegrating vector, and abstracts from the explicit CCI spline. The present paper complements MARTIN by providing a freely estimated benchmark against which calibrated coefficients can be evaluated, and by surfacing the identification choices that drive the estimated speed of adjustment.

A separate strand of Australian work has examined the cyclical co-movement of consumption with credit and housing conditions through reduced-form approaches: Bayesian VAR frameworks, RBA Bulletin analyses of mortgage payments and household debt burdens, and the housing-leverage and household-balance-sheet treatments in the broader RBA Research Discussion Paper series. These approaches identify shorter-run dynamics and stress-test sensitivities but do not deliver the long-run cointegrating vector that the LIVES specification produces, and so are complementary to the present analysis rather than substitutes for it.

1.3.4 2.4 Identification of credit conditions

The credit-conditions index is the most contested ingredient in the LIVES framework. Williams (2010) constructs CCI as a latent variable identified by smoothed-step dummies (SDMMA series — five-quarter moving averages of four-quarter moving averages of step dummies) at four institutional turning points: 1979 (Campbell Committee, removal of interest-rate ceilings on bank deposits), 1992 (banking distress and the entry of the first mortgage originator, Aussie Home Loans), 1998 (the rise of non-bank financial institutions and securitisation), and 2007 (the global financial crisis tightening). The institutional chronology of Australian financial deregulation underpinning these choices is documented in Battellino and McMillan (1989) and Edey and Gray (1996); Bayoumi (1993) provides a cross-country analysis of the consumption response to financial liberalisation, including Australia, that quantitatively validates a structural CCI shift in the early 1980s.

In the system-estimation context of Muellbauer and Williams (2012), each spline coefficient is identified by being a common factor across the four equations: the same CCI value enters consumption, house prices, the mortgage stock and home equity withdrawal, with different loadings. This common-factor identification is the central methodological contribution of the LIVES family. In single-equation OLS, the identification is weaker but the spline can still be estimated under sign-prior restrictions enforced by general-to-specific reduction (Hendry-Krolzig 2005). We adopt this approach as a robustness check (Spec 8) in the present paper.

The alternative, observable proxy for CCI — the ratio of housing credit flow to disposable income, in logs — has the advantage of being measured directly but the disadvantage of being available only from the early 2000s, after the most informative deregulation episodes have already occurred. We use the observable proxy as a short-run regressor in the specifications that include it (Specs 2 and 5), and the spline approach for the long-run identification in Spec 8.

A separate identification approach uses survey-based credit-conditions indices. The Federal Reserve's Senior Loan Officer Opinion Survey is the prototype; the European Central Bank's Bank Lending Survey serves the same function in the euro area. The Reserve Bank of Australia operates a qualitative Liaison Programme but does not publish a long-running numerical index of credit conditions for households. The unavailability of such a survey-based Australian series is the practical reason for adopting Williams' spline approach in the LIVES tradition.

1.3.5 2.5 Permanent-income forecasting

The empirical operationalisation of permanent income — the discounted expected weighted average of future income — requires either explicit forecasts or a parametric assumption about the income process. The standard practice in the consumption literature has been to assume an AR(p) process for log income, fit it on the available sample, and aggregate the multi-step-ahead forecasts using exponentially declining weights with discount factor δ . This recipe

descends directly from the PIH literature of Hall (1978) and the consumption-Euler-equation tests of Campbell and Mankiw (1989).

Two recent methodological developments have improved on this recipe. Jordà (2005), originally proposed for impulse-response identification, estimates the discounted weighted average of income over the relevant horizon directly as the dependent variable in a single regression, with predictors observable at time t . The local-projection approach sidesteps the compounding of AR misspecification across forecast horizons to which the standard recipe is vulnerable, and admits a richer predictor set than is feasible in a parsimonious AR(p). Carroll, Slacalek and Tokuoka (2014), in ECB Working Paper 1648, document substantial differences between the permanent-income series implied by AR(p) forecasting and by direct local-projection forecasting in the consumer-expenditure context. The differences are concentrated at structural-break episodes, where AR(p) forecasts inherit the slow adjustment of the estimated AR coefficients while local projections can incorporate predictors capturing the regime change.

De Bonis et al. (2024) explicitly adopt the Jordà local-projection approach for their Italian permanent-income series, reporting that the choice "captures much of the slow-down of permanent income growth in the early 1990s" — a structural feature of Italian growth that the AR-based forecaster missed. The Italian local projection includes a $\log(\text{labour_force} / \text{population})$ predictor, which slowly trends with demographic change and is a natural input to long-horizon income forecasting. We adopt the same predictor in our Italy-style PI helper (§4.3) and find a quantitatively similar role: the PI series implied by local projection in Australia diverges materially from the AR-based PI series in the early 1990s and after the 2008 GFC. Substantively, the implied long-run coefficient on $\log(y^p/y)$ flips from significantly negative under the AR forecaster to significantly positive under the local-projection forecaster, matching Williams' calibrated value.

1.3.6 2.6 Where this paper sits

The contribution of this paper is best understood as a contemporary single-equation revisit and extension of Williams (2010) with three methodological refinements drawn from the parallel Italian and French implementations. First, we estimate on a non-overlapping post-1988 sample that adds sixteen additional years of post-GFC data, providing external validation of Williams' published wealth-coefficient estimates on data he did not see. Second, we introduce three corrected data inputs — RBA F6 mortgage rate (replacing a previously biased ABS-implicit fallback), ABS A84423091W 15+ population (replacing a previously biased single-year-of-age cohort sum), and ABS Cat 5206.0 Table 20 income components (enabling Williams' 2009 non-property-income recipe) — that are typical of the revisions a current vintage of the same data would deliver. Third, we adopt the Italian methodology of single-equation OLS with a structured robustness suite covering instrumental variables, joint SUR estimation, multi-window Chow tests, scaled-income and Drehmann real-rate alternatives, and a Williams-style spline credit-conditions robustness column.

The headline paper does not estimate the multi-equation LIVES system itself, though we scaffold the multi-equation build in a separate companion directory. Williams (2010) estimates four equations jointly by FIML; the headline of the present paper estimates the consumption equation alone. The Italian experience (De Bonis et al. 2024, §4.2) suggests that single-equation OLS produces consumption-equation coefficients "only a whisker away" from joint SUR

estimation, and we replicate that finding (§8.2). On our back-extended sample a two-equation consumption + house-prices SUR finds residual correlation $\hat{\rho} \approx 0.0007$ — joint estimation gives no efficiency gain at the quarterly frequency — so the case for the full multi-equation build rests on cross-equation parameter restrictions rather than on efficiency. The full multi-equation extension is left to a companion paper, with the scaffolding documented separately.

The paper does extend the sample back to **1976Q3**. The public-data backbone consists of a Treasury TRYM long-run house-price series (1959Q3+), the RBA D03 M3 monetary aggregate (1959Q3+), the RBA D02 total-credit splice (1976Q3+), and a historical labour-force compilation (1964Q3+). For the 1976Q3–1988Q2 window where ABS sectoral household balance-sheet data are unavailable, we construct aggregate and disaggregated wealth proxies anchored at 1988Q3 (§3.9–§3.13). The empirical finding from refitting the disaggregated no-CCI specification on the back-extended sample (§7.3, §8.15, §9) is that the residual structural-coefficient gap with Williams does *not* close on the longer window — λ moves closer but individual wealth coefficients fall further from Williams' values — which we read as evidence that the binding constraint is the single-equation framing itself rather than the post-1988 sample window.

1.4 3. Data and measurement

The dataset assembles quarterly Australian macroeconomic and household-sector observations from **1976Q3 to 2024Q4 (n = 194)**. The public-data backbone for the pre-1980 window is built from a Treasury TRYM long-run house-price series, the RBA D03 M3 monetary aggregate, the RBA D02 total-credit splice, and a historical labour-force compilation. The disaggregated wealth components remain bounded at 1988Q3 by their primary source (ABS Cat 5232.0 Household Balance Sheet); for the 1976Q3–1988Q2 window we construct proxies (§3.13 below) that growth-rate-splice each component onto its 1988Q3 official value via the most relevant available aggregate.

Estimation is performed on the largest contiguous subset for which all variables in a given specification are observed:

- **Spec 1–3 (aggregate net worth)** with the back-extension proxy fits on n=190 (1977Q3–2024Q4; the binding constraint is `real_rate`, which needs a 4-quarter CPI lag for inflation computation). On the official `networth_y` (1988Q3+ ABS) it fits on n=146.
- **Spec 4–7 (disaggregated wealth)** with the back-extension proxies fit on n=190; on the official disaggregated series, n=146.
- **Spec 6 (preferred — disagg + post-2008 PI shift + SR dynamics)** is bounded at 2002Q3+ regardless of the wealth-component proxy because its short-run CCI variable (`d2_logcci_lag2`) depends on `cci_ratio = log(housing_loan_flow / income)` from ABS Cat 5601.0, which only starts 2002Q3. n=86 in both samples.
- **Spec 8 (Williams CCI interactions)** fits on n=146 (bounded by disaggregated wealth).
- **Spec 9 (Kalman state-space CCI)** fits on the same window as the disaggregated specs; the Kalman extraction itself is a separate state-space step.
- **Spec 10 (Williams-prior calibrated)** fits via iterative fixed-point OLS on n=146.

1.4.1 3.1 Aggregate consumption and income

Real per capita consumption (`cons_real_pc`) is constructed from ABS Cat 5206.0 Table 8 (Household Final Consumption Expenditure, chain volume measures, seasonally adjusted)

divided by the civilian population aged 15 years and over (ABS series A84423091W, monthly, averaged to quarterly arithmetic mean; sample 1978Q2–2024Q4, sourced directly from the ABS historical series workbook). Following Williams (2010, 2009), we do not sum the single-year-of-age cohorts in ABS Cat 3101.0; in current ABS vintages, the Persons series only extends to age 47 in the disaggregated file, producing a population total approximately 35 per cent below the true Estimated Resident Population.

Real per capita household disposable income (ydi_real_pc) is the quarterly seasonally adjusted nominal series from ABS Cat 5206.0 Table 20 (Household Income Account), deflated by the consumption deflator implied by Tables 8 (chain volume) and 8 (current prices), and divided by the same 15+ population denominator. Following standard practice (Blinder-Deaton 1985), and as the headline empirical specification, we use gross disposable income; we report a non-property income (NPY) alternative constructed per Williams (2009) §4.2.1 in §3.6 below.

1.4.2 3.2 Household balance sheet

Household sector balance sheet stocks are sourced from ABS Cat 5232.0 Table 35 (Household Balance Sheet Aggregates, current prices, \$ billion), quarterly from 1988Q3:

- Currency and deposits ($fin_deposits$)
- Shares and other equity ($fin_equities$)
- Superannuation reserves (fin_super)
- Total household liabilities (loans and placements; fin_loans)
- Residential land and dwellings ($housing_wealth$)
- Closing net worth (ABS series A83722648X)

All balance-sheet stocks are converted to real values by dividing by the normalised consumption deflator and to per-capita-15+ values by dividing by the 15+ population.

The wealth-to-annualised-income ratios used in the long-run consumption equation are constructed as:

- $ha_y = housing\ wealth / (4 \times \text{quarterly nominal disposable income})$
- $eq_y = equities\ (ex\text{-}super) / (4 \times \text{quarterly nominal disposable income})$
- $super_y = superannuation\ reserves / (4 \times \text{quarterly nominal disposable income})$
- $nla_y = (deposits - \text{total household debt}) / (4 \times \text{quarterly nominal disposable income})$
- $debt_y = \text{total household debt} / (4 \times \text{quarterly nominal disposable income})$
- $networth_y = \text{closing net worth} / (4 \times \text{quarterly nominal disposable income})$

We adopt the Italian convention (De Bonis et al. 2024, eq. 2.5; Table 3 column 3) of defining net liquid assets as **deposits net of total household debt**, with the cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ imposed implicitly by construction. We test this restriction formally in §8 below; it is accepted at the 5 per cent level in every disaggregated specification and sample window.

1.4.3 3.3 Mortgage interest rate

The nominal mortgage rate ($mortgage_rate$) is the RBA Standard Variable Owner-Occupier Rate (RBA Bulletin Table F5, series FILRHLBVS), monthly from January 1959, averaged to quarterly. We source the historical series from the published RBA archive rather than the live

readrba API, to ensure a stable vintage; the rate peaks at 17.0 per cent in 1989Q3, consistent with the Hawke-Keating recession.

The real mortgage rate (`real_rate`) is the nominal rate less the 4-quarter-ended percentage change in the consumption deflator. We do not adopt the Italian Drehmann (2017) amortising-mortgage adjustment in the preferred specification but report it as a robustness check.

1.4.4 3.4 House prices

The headline house price index (`hpi`) is constructed by chain-linking **four** sources, with the deepest segment now extending to 1959Q3:

Layer	Source	Coverage	Splice convention
TRYM Model historical database	Treasury Macroeconomic (house_price_history_long.csv)	1959Q3–2018Q2	growth-rate, anchored at 1986Q2
Legacy	houseprice_old.csv (privately compiled dwelling-price index, monthly→quarterly)	1986Q2–2003Q3 pre-2003 index,	growth-rate, anchored at 2003Q4
Bridge	ABS Cat 6416.0 Residential Property Price Index, 8-capital-cities ("old method")	2003Q4–2017Q2	growth-rate, anchored at first overlap
Current	ABS Cat 6432.0 Total Value of Dwellings, mean price	2003Q3–2024Q4	(the modern overlay)

The TRYM source supersedes the BIS Shrapnel / REIA chain used in Williams (2010): the TRYM historical compilation already incorporates the same BIS Shrapnel (pre-1978), REIA (1978–1986) and ABS (post-1986) segments that Williams used, pre-chained into a single coherent 235-quarter series. The earliest binding observation for `hpi` is therefore now **1959Q3**, three years deeper than even Williams' fullest sample.

1.4.4.1 3.4.1 Splice methodology — pure growth-rate chain-linking

For each adjacent pair of layers, the splice anchors the level at the first quarter where both series are non-NA, then back-casts via the base series' own QoQ growth rates:

```

chained[t] = overlay[t_anchor] × (base[t] / base[t_anchor])  for t < t_anchor
chained[t] = overlay[t]                                       for t >=
t_anchor

```

This standard ABS chain-linking convention preserves the base series' growth rates exactly while pinning the level to the overlay at the join. By construction there is no level discontinuity at any join quarter. An earlier implementation used `mean(overlay/base)` over the full overlap, which produced step jumps at join quarters where the overlap ratio drifts (notably a -17% step at 1986Q2 under the long TRYM↔legacy overlap and a $+10\%$ step at 2011Q2→Q3 under the bridge↔current overlap with mismatched units — the bridge is an index while current is a \$-value mean). Both artefacts were eliminated by the growth-rate convention.

The relative house-price-to-income ratio used in estimation is

$\ln_hp_over_y = \log(hpi / (ydi_ann_nom / pop_millions / (cons_deflator_norm / 100)))$

This is the log of the real house-price index divided by real disposable income per capita.

1.4.5 3.5 Credit conditions index

The credit conditions index (CCI) is the most contested input. Williams (2010, 2012) constructs CCI as the latent factor identifying credit liberalisation episodes: a spline of smoothed-step dummies at the institutional turning points 1979 (Campbell Committee, end of interest ceilings), 1992 (banking distress, Aussie Home Loans), 1998 (NBFI expansion), and 2007 (GFC), estimated jointly with the consumption, house-price, mortgage-stock and home-equity-withdrawal equations under sign priors (positive 1979–1990, negative 1992–1994, positive 1998–2006, negative 2007 onwards).

We adopt a hybrid:

- **Default observable proxy** (`cci_ratio`): the log of housing credit flow (ABS Cat 5601.0 New Loan Commitments Value, total housing) divided by the eight-quarter moving average of nominal disposable income. Available 2002Q3 onward ($n=90$); used as a short-run regressor in specifications 2 and 5. We do not back-extend this with a mortgage-spread proxy; this option (the default in earlier vintages of the codebase) is retained behind a feature flag but disabled.
- **Williams-style 4-knot spline** (`cci_williams`, optional via `USE_INSTITUTIONAL_CCI = TRUE`): smoothed-step dummies at 1979Q1, 1992Q1, 1998Q1, 2007Q1 estimated inside the disaggregated consumption equation by general-to-specific reduction with sign priors enforced by drop-on-violation (in the spirit of Hendry-Krolzig 2005). On our 1988Q4-onwards sample, two of the four knots survive: 1998 with the expected positive coefficient and 2007 with the expected negative coefficient; the 1979 knot is constant within our window and the 1992 knot violates its prior. We discuss the implications in §5 below.

The first-home-buyer share (`fhb_share = fhb_loans / total_new_loans`) is constructed from ABS Cat 5601.0 from 2002Q3.

1.4.6 3.6 Williams (2009) non-property income

The Williams (2009) §4.2.1 non-property income measure (`npv_real_pc`) adjusts gross disposable income by removing imputed property income and a corresponding share of property-related taxation. Following Williams (2009 p. 10):

```
npv_rec = total_income_rec - GOS_dwellings - prop_inc_rec
property_tax_share = (GOS_dwellings + prop_inc_rec) / total_income_rec
npv_pay = total_income_pay - prop_inc_pay - property_tax_share × income_tax_payable
NPY = npv_rec - npv_pay
```

The four input components are sourced from ABS Cat 5206.0 Table 20: compensation of employees, gross operating surplus on dwellings, property income receivable, social assistance benefits, property income payable, total income receivable, total income payable, and income tax payable.

The implied non-property income share of disposable income averages 0.84 over 2010–2024, consistent with Williams' implicit ~0.85 weighting. We use NPY as a robustness column (see §8) and report sensitivity to the income measure as a key methodological gap that explains

approximately 30 per cent of the speed-of-adjustment divergence between our preferred specification and Williams.

1.4.7 3.7 Demographics and dummies

The prime-working-age share (`prime_age_share` = age 25-54 share of total ERP) is constructed from ABS Cat 3101.0 single-year-of-age cohorts (Male + Female, summed; ratios are robust to the truncation discussed in §3.1). Annual data are interpolated to quarterly via cubic spline.

Five Australia-specific narrative dummies enter the default dummy set:

- `d_neg_gearing_8587`: 1985Q3–1987Q3 negative-gearing tax restriction
- `d_recession_1991`: 1991Q2 ("recession we had to have")
- `d_apra_2014`: 2014Q4 macroprudential investor-loan-growth cap (smoothed-step ogive)
- `d_apra_2017`: 2017Q2 macroprudential interest-only-loan cap (smoothed-step ogive)
- `d_jobkeeper_2020`: 2020Q2–2021Q1 JobKeeper income support

Together with the standard four (`d2000_gst`, `d2008_gfc`, `d2020_covid`, `d2020_rebound`), these constitute the full dummy set; zero-variance dummies are silently dropped per specification.

1.4.8 3.8 Coverage and reproducibility

The full reproducibility kit (R 4.5.3, renv-pinned dependencies, raw ABS workbooks, project-supplied CSVs, master quarterly dataset as CSV and RDS, full estimation pipeline with three execution modes, GitHub Actions CI) accompanies this paper. The master dataset has **194 quarters** × **~110 columns** and is available as a portable CSV (`master_data.csv`) for hand-editing or off-line replay. See appendix on data construction for each variable's source identifier, vintage, and splicing recipe.

1.4.9 3.9 RBA D-tables

Three RBA historical statistical tables are used to support the pre-1988 portion of the sample.

1.4.9.1 3.9.1 RBA D03 — Monetary aggregates (M3)

The M3 monetary aggregate (`m3_aggregate`) is loaded from `d03hist.xlsx`, series `DMAM3N` (M3, original/not-seasonally-adjusted, \$ billion). M3 is total economy-wide currency plus transaction deposits plus all other deposits at ADIs, plus certificates of deposit issued by ADIs. Coverage: monthly, **1959Q3–2026Q1** (continuous, no series breaks). Aggregated to quarterly by mean of the three monthly observations.

M3 is the headline liquid-asset proxy for the pre-1988 portion of the sample where ABS sectoral household-balance-sheet deposits aren't available. Williams (2010) used the same M3 series, multiplied by the household share of factor income, for his pre-1988 splice.

1.4.9.2 3.9.2 RBA D02 — Lending and credit aggregates (total credit)

Total credit (`credit_total_d02`) is constructed by growth-rate splicing two RBA D02 series across the July 2019 RBA conceptual reform:

- `DLCACN` (Total credit, original, \$ billion) for 1976Q3–2019Q2
- `DLCACSFN` (Total credit including select financial businesses, the post-2019 successor) for 2019Q3+

The two series have no quarterly overlap (`DLCACN` ends 2019-06, `DLCACSFN` starts 2019-07). The splice anchors levels at the boundary (`first_post` × `pre[t]` / `last_pre`) so the join

is continuous in level; the implicit growth rate at the boundary is exactly zero (no overlap to estimate it from). For analyses that hinge on the 2019Q2→Q3 quarter specifically, treat with caution. Coverage 1976Q3+.

This is total credit, not housing-specific. The housing-specific series in D02 (DLCACOHN, DLCACIHN) only goes back to 1990Q1 in current vintage and so cannot extend the housing-credit history pre-1990.

1.4.9.3 3.9.3 RBA D01 – Growth in selected financial aggregates

Downloaded as `d01hist.xlsx` (provides monthly growth rates of the same aggregates whose levels are in D02). Williams (2010) used D01 housing-credit growth rates to back-cast D02 levels pre-1976; in the current vintage D02 already extends to 1976Q3 so D01 is not strictly needed for back-extension. Retained for future use.

1.4.10 3.10 Pre-1978 labour force (`labour_force_historic.csv`)

A user-supplied CSV (`labour_force_historic.csv`, 188 quarterly rows 1964Q3–2011Q2) provides:

Column	Definition	Units
<code>pop_total</code>	Total resident population (annual + interpolation)	thousands
<code>pop_15_64</code>	Working-age population (15–64)	thousands
<code>labour_force</code>	Civilian labour force	thousands
<code>unemployed</code>	Unemployed persons (level)	thousands

Provenance: the source compilation pulls together pre-1978 series from ABS Cat 6204.0 (historical labour force, archived), ABS Year Book Australia, the Foster (1996) *Australian Economic Statistics 1949–50 to 1996–97* compilation, and RBA Occasional Paper No 8. Conceptually consistent with the current ABS Cat 6202.0 series (which begins Feb 1978) at the join.

The historic series are growth-rate spliced onto the modern (1978+) series at 1978Q1:

- `pop_15_64` → `pop_millions` (the working-age population denominator used in per-capita normalisation)
- `labour_force` → `master labour_force`
- `unemployed/labour_force × 100` → `master unemp_rate` (level replacement before 1978Q1)

Side-effect of this splice: the six quarters 1976Q3–1977Q4 now have non-NA values for all per-capita and labour-force-derived variables (`cons_real_pc`, `ydi_real_pc`, `npv_real_pc`, `lf_share`, etc.). On the data download, `cons_real`, `ydi_nom`, `mortgage_rate`, `hpi`, `M3`, `total credit`, and `prime_age_share` all extend to 1976Q3 already; the labour-force splice was the binding remaining constraint.

`pop_total` is also exposed in the master as `pop_total_thousands` (historic-only 1964Q3–2011Q2 because there's no modern 1978+ counterpart in ABS 6202; ABS Cat 3101.0 publishes it but on a different release schedule). It can be used as an alternative per-capita denominator (total resident vs civilian 15+) for sensitivity analyses.

1.4.11 3.11 Household-allocated M3 (m3_household_proxy)

The pre-1988 liquid-asset proxy is constructed by allocating M3 to the household sector via the wage share of GDP:

$$m3_household_proxy = m3_aggregate \times wage_share / 100$$

Williams (2010) uses the household factor income share for this allocation. The wage share alone (compensation of employees / GDP) is a defensible approximation: it captures most of household factor income (wages dominate Australian household income; mixed income + property income receivable add ~10 percentage points and track wage share over time). A documented simplification for our phase-1 back-extension; for a fully Williams-faithful splice, replace wage_share with the broader (compensation + mixed income + property income receivable) / GDP series.

wage_share itself is loaded from a user-supplied CSV (household_income.csv) which extracts ABS Cat 5206.0 Table 24 analytical series A2302604K (compensation of employees, % of GDP), 1959Q3–2024Q4. Range over the sample: 49–62%, with values ~60% in the 1970s, falling to ~50% today (capital-share rise).

Coverage of m3_household_proxy: 1976Q3+ (limited by wage_share in the supplied CSV), values \$22 billion (1976Q3) to \$1,673 billion (2024Q4).

1.4.12 3.12 Aggregate net-worth proxy (networth_y_proxy)

To enable Spec 1–3 fits on the back-extended sample, we construct an aggregate net-worth proxy that:

1. Uses M3-allocated-to-households (m3_household_proxy) plus a hpi × pop_millions back-cast of housing wealth as the raw wealth aggregate.
2. Growth-rate-splices that raw aggregate ratio onto the official networth_y at 1988Q3 (so the proxy equals the official series from 1988Q3 onwards and back-casts smoothly through 1976Q3).

The housing-wealth back-cast anchors at the first available official housing_wealth observation (1988Q3) and back-casts via housing_wealth × hpi × pop_millions growth, holding dwellings per capita constant:

$$\begin{aligned} \text{housing_wealth_proxy}[t] &= \text{housing_wealth}[1988Q3] \\ &\quad \times (\text{hpi}[t] / \text{hpi}[1988Q3]) \\ &\quad \times (\text{pop_millions}[t] / \text{pop_millions}[1988Q3]) \\ &\quad \text{for } t < 1988Q3 \\ &= \text{housing_wealth}[t] \text{ for } t \geq 1988Q3 \end{aligned}$$

The aggregate proxy is then:

$$\begin{aligned} \text{raw_proxy}[t] &= (m3_household_proxy[t] + \text{housing_wealth_proxy}[t]) / \\ \text{ydi_ann_nom}[t] & \\ \text{scale} &= \text{networth_y}[1988Q3] / \text{raw_proxy}[1988Q3] \quad (\approx 1.90 \text{ in current} \\ &\quad \text{vintage}) \\ \text{networth_y_proxy}[t] & \\ &= \text{networth_y}[t] \quad \text{for } t \geq 1988Q3 \\ &= \text{raw_proxy}[t] \times \text{scale} \quad \text{for } t < 1988Q3 \end{aligned}$$

Caveats explicitly documented in data.md §3.4b:

- The back-cast omits equities and super (quantitatively small in 1976–1988 — Australian super pre-Superannuation Guarantee 1992 was a negligible household asset class).
- The back-cast omits debt netting (mortgage debt was a much smaller share of household balance sheets in the 1970s than today).
- Use only for back-extension exercises; never as a substitute for `networth_y` on the modern sample where the official series exists.

The proxy values across key dates: $5.09 \times$ annual income (1976Q3), 4.67 (1980Q1), 4.72 (1985Q1), 5.37 (1988Q3 — anchored to official), 10.23 (2024Q4). Shape is consistent with the historical Australian wealth-to-income trend (flat through 1970s/early 80s, sharply rising post-1985 financial deregulation).

1.4.13 3.13 Disaggregated wealth proxies

Four additional proxies extend the disaggregated wealth components to the back-extended 1976Q3+ sample. Each equals the official series for $t \geq 1988Q3$ by construction; for $t < 1988Q3$ it back-casts via the most relevant available aggregate.

ha_y_proxy = `housing_wealth_proxy / ydi_ann_nom`. Uses the `hpi`×`pop` back-cast described in §3.12. Values: 2.68 (1976Q3) → 2.83 (1988Q3) → 6.41 (2024Q4).

fin_deposits_proxy anchors `fin_deposits[1988Q3]` and grows by `m3_household_proxy`:

```
fin_deposits_proxy[t] = fin_deposits[1988Q3]
                      × m3_household_proxy[t] / m3_household_proxy[1988Q3]
                      for t < 1988Q3
```

fin_loans_proxy anchors `fin_loans[1988Q3]` and grows by RBA total credit:

```
fin_loans_proxy[t] = fin_loans[1988Q3]
                   × credit_total_d02[t] / credit_total_d02[1988Q3]
                   for t < 1988Q3
```

nla_y_proxy = $(\text{fin_deposits_proxy} - \text{fin_loans_proxy}) / \text{ydi_ann_nom}$. Values: +0.20 (1976Q3, households are net liquid creditors) → -0.05 (1988Q3) → -0.72 (2024Q4, modern net debtor position). The sign-flip around 1988 captures the post-deregulation debt build-up.

eq_y_proxy: held constant at the 1988Q3 value pre-1988 (Option B in our methodology — Australian household equity holdings were a small wealth share in the late 1970s/early 80s; the constant assumption introduces little level error and is straightforward to upgrade to ASX-All-Ords back-cast). Value: 0.60 (constant) → 0.92 (2024Q4).

super_y_proxy: linear ramp from $0.1 \times$ the 1988Q3 value at 1976Q3 to the 1988Q3 value, then official thereafter. The 0.1 anchor matches Williams (2010) Table A.1 ballpark for the pre-Superannuation-Guarantee era (SGC mandate 1992). Values: 0.07 (1976Q3) → 0.66 (1988Q3) → 2.44 (2024Q4).

1.4.13.1 3.13.1 Coherence check — disagg sum vs aggregate proxy

Both `networth_y_proxy` (M3-allocated + housing back-cast, growth-rate-spliced onto the official broad `networth`) and the **sum-of-disaggregated** `networth_y_disagg_proxy` = `ha_y_proxy` + `nla_y_proxy` + `eq_y_proxy` + `super_y_proxy` are exposed in the master.

At 1988Q3 (the boundary):

- official `networth_y` (broad, uses ABS A83722648X closing net worth): **5.37**

- `networth_y_proxy` (aggregate, anchored to broad): **5.37** by construction
- `networth_y_disagg_proxy` (narrow, sum of components): **4.04**

The 25 percent gap between aggregate and disagg sum is the "other wealth" component of ABS closing net worth (life office reserves, unincorporated business equity, etc.) that is absent from the narrow definition. For Spec 4–7 fits, which use the disaggregated components individually, the disagg sum is the implicit reference. For Spec 1–3 fits, the aggregate proxy is used.

1.4.14 3.14 Master variable coverage tiers (under 1976Q3+ spine)

After all splicing, master variable coverage falls into the following tiers:

First non-NA	n	Variables (selected)
1976Q3	56	cons, ydi, hpi, mortgage_rate, M3, total credit, prime_age_share, all dummies, m3_household_proxy, ha_y_proxy, nla_y_proxy, eq_y_proxy, super_y_proxy, networth_y_proxy, ln_networth_y_proxy, cons_real_pc, ydi_real_pc, npy_real_pc, labour_force, unemp_rate, lf_share, pop_millions, ln_cons_real_pc, ln_ydi_real_pc, ln_hp_over_y, ecm_lag (1976Q4)
1977Q3	2	real_rate, hicp_4q_ann (4-quarter CPI lag)
1978Q2	1	ydi_ann_8qma (8-quarter MA)
1988Q3	21	OFFICIAL disaggregated wealth (ha_y, eq_y, super_y, nla_y, ifa_y, networth_y, debt_y), housing_wealth, fin_deposits/equities/super/loans (RAW from ABS 5232)
2002Q3	5	cci_ratio, fhb_share, housing_loan_flow, fhb_loans, non_fhb_loans
2009Q1	2	mortgage_interest_burden_rba, mortgage_payment_burden_rba (RBA E13)

1.5 4. Model

1.5.1 4.1 Functional form

We estimate the standard Muellbauer error-correction consumption equation adapted to Australian data following Williams (2010) eq. (7):

$$\Delta \ln c_t = \lambda [\alpha_0 + \gamma_{HA} \cdot (HA/y)_{t-1} + \gamma_{IFA} \cdot (IFA/y)_{t-1} + \gamma_{NLA} \cdot (NLA/y)_{t-1} + \gamma_{HP} \cdot \ln(p^h/y)_{t-1} + \alpha_r \cdot r_t + \phi \cdot \ln(y^p/y)_t + ecm_lag_t] + \sum \beta_j Z_{jt} + \sum \delta_k D_{kt} + \varepsilon_t$$

where

- c_t is real per capita household consumption
- y_t is real per capita disposable income
- (HA/y) , (IFA/y) , (NLA/y) are the housing, illiquid financial and net liquid wealth-to-annualised-income ratios
- p^h_t is the real house price index, so $\ln(p^h/y)$ is the affordability/down-payment ratio
- r_t is the ex post real mortgage rate

- y^p_t is the discounted weighted average of expected future real per capita income (the permanent-income concept)
- $ecm_lag_t = \ln(c_{t-1}) - \ln(y_t)$ is the canonical Engle-Granger error-correction term
- Z_{jt} are short-run dynamic regressors and D_{kt} are narrative dummies as documented in §3.7
- $\lambda < 0$ is the speed of adjustment and α_0 , γ^* , α_r , ϕ are long-run cointegrating coefficients (the structural parameters in the LIVES literature)

The structural γ coefficients are recovered as $OLS_coef / |\lambda|$ from the OLS regression of $\Delta \ln c$ on the right-hand side; we report both forms throughout (see §7).

1.5.2 4.2 Sign priors

Theoretical sign priors on the long-run coefficients are:

- $\gamma_{HA} \geq 0$ (housing wealth: collateral and lifetime-income channels)
- $\gamma_{IFA} \geq 0$ (illiquid financial wealth)
- $\gamma_{NLA} \geq 0$ (net liquid: buffer-stock + intertemporal substitution)
- $\gamma_{HP} \leq 0$ at credit-tight regimes (down-payment penalty); sign ambiguous at credit-loose regimes (collateral channel dominates; Williams 2010 reports positive coefficient at the 2007 CCI peak)
- $\alpha_r \leq 0$ (intertemporal substitution)
- $\phi \in [0, 1]$ (Hall 1978 PIH; calibrated by Williams)
- $\lambda < 0$ (stable error correction)

We use these as informal screens (§6.2) rather than imposing them as formal Bayesian priors; freely estimated coefficients allow us to report violations of the priors as substantive findings rather than artefacts of imposed restrictions.

1.5.3 4.3 Permanent-income forecasting

Permanent income y^p_t is the discounted weighted average of expected log income over a 40-quarter horizon at quarterly discount factor $\delta_q = 0.95^{(1/4)}$:

$$\ln(y^p_t / y_t) = E_t \left[\sum_{h=1}^{40} w_h \ln(y_{t+h}) \right] - \ln(y_t) \text{ where } w_h = \delta_q^{(h-1)} / \sum_{h=1}^{40} \delta_q^{(h-1)}$$

We implement two forecasting methods. The canonical method for the WP results below is **Italy LP**; **AR** is reported as a methodology robustness column in §8.9.

- **Method 'Italy' (canonical)**: Jordà (2005) local projection. For each t where the future horizon is observable, the discounted weighted average is computed directly as the dependent variable in a single regression on a richer predictor set including $\log(lf_share)$ (the Italian innovation, capturing slow-moving demographic effects on trend income), trend, post-2008 split-trend, 4-quarter-MA log income, unemployment rate, and 4-quarter-difference dynamics. Forecast values are then constructed as fitted values of this single regression for every t .
- **Method 'AR' (robustness)**: rolling AR(8) regression of log income on eight own lags plus a linear trend, post-2008Q3 step dummy, and trend-break interaction. Forecasts are aggregated over 40 horizons using the discount weights. Optional predictors `unemp_rate`,

`log_oil`, `log_reer`, `log_stocks` are added if available. A 2008Q3 ogive learning weight smoothly attenuates the term over 15 quarters to a steady-state weight of 0.5.

The two methods differ materially on two coefficients in the consumption equation (see §8.9): the speed of adjustment and the long-run permanent-income coefficient. Canonical Italy LP delivers $\lambda = -0.180$ in the preferred specification, against approximately -0.05 under AR. We report both, with Italy LP carrying the headline narrative.

1.5.4 4.4 Wealth definition

Following Italy (De Bonis et al. 2024 §2 eq. 2.5; Table 3 col. 3) and Williams (2010), net liquid assets are defined as

$$\text{NLA} = \text{liquid_assets} - \text{total_household_debt}$$

and we test the cross-equation restriction $\gamma_{\text{LA}} + \gamma_{\text{LOANS}} = 0$ by refitting each disaggregated specification with deposits and debt entered separately (`nla_y_unrestricted` and `debt_y`) and conducting a Wald test of equality with opposite signs. Italy formally tests and accepts this restriction; we replicate the test on Australian data in §8 (`australia_nla_restriction_test.csv`).

1.6 5. Identification of credit conditions

This section is the methodological pivot of the paper. The credit conditions index CCI is jointly determined with consumption, house prices, mortgage debt and home equity withdrawal in the LIVES framework (Muellbauer-Williams 2012). In a single-equation setting, its identification rests on either an observable proxy or institutional prior knowledge.

1.6.1 5.1 The Williams smoothed-step spline approach

Williams (2010) constructs CCI as a spline of SDMMMA smoothed-step dummies — a 5-quarter moving average of a 4-quarter moving average of a 0/1 step — at institutional turning points in the Australian financial-policy chronology. Each knot's coefficient is constrained by a sign prior derived from institutional history (deregulation episodes positive; retrenchment episodes negative), enforced by Hendry-Krolzig (2005) drop-on-violation general-to-specific reduction.

Williams' canonical paper uses four knots: 1979Q1 (Campbell Committee, removal of interest rate ceilings), 1992Q1 (NBFI distress post the early-1990s recession), 1998Q1 (NBFI/securitisation expansion), and 2007Q1 (GFC retrenchment). The four-knot choice reflects the institutional information available at the time of his 1977-2008 sample: STAMP-derived turning points and a deregulation calendar ending shortly after the GFC.

On our 1988Q4-onwards sample, only one of Williams' four canonical knots survives sign-prior reduction. A direct replication of the Williams 4-knot specification yields:

Williams knot	Sign prior	Status on 1988+ sample
1979Q1	+	aliased (constant within window)
1992Q1	–	sign violator (data signal +ve)
1998Q1	+	sign violator (data signal –ve)
2007Q1	–	survives, coef ≈ -0.014

The 1979 deregulation knot is mechanically uninformative because the smoothed step reaches unity by 1980Q2, three years before our window opens. The 1992 and 1998 knots fail their institutional sign priors: the post-1988 sample observes the recovery from the early-1990s banking distress (during which credit growth resumed and the OLS coefficient turns positive) and the late-1990s NBF1 period without the contrast against the prior tight regime that identifies the loosening direction.

A direct 4-knot replication is therefore *not* identifying the four distinct credit-conditions episodes Williams' framework attributes to the spline. It is identifying one – the 2007 GFC tightening – plus a constant.

1.6.1.1 5.1.1 The maximal-GETS Australian CCI

Rather than impose Williams' published knot count on a sample that cannot identify three of his four knots, we adopt a **maximal-GETS approach**: starting from a richer 15-knot candidate set covering the documented Australian financial-policy chronology, we let drop-on-violation reduction prune knots that are aliased or violate their institutional sign prior. The candidate institutional events span Campbell '79, the housing-finance deregulation of '86, state-bank distress '90, banking distress '92/'93, the Wallis report and the establishment of APRA in '98, the GFC tightening of '07, the deposit-guarantee scheme '08, the FHB Boost '09, the APRA macroprudential rounds of '14 and '17, the Hayne Royal Commission of '19, the APRA cap removal and buffer reduction of '19Q3, the COVID/JobKeeper episode of '20, and the APRA buffer hike of '21.

On the 1988Q4-2024Q4 sample this candidate set yields six surviving knots:

Knot	Sign prior	Coef (OLS)	Reading
1992Q1	–	–0.020	Banking distress / Aussie Home Loans
2007Q3	–	–0.007	GFC tightening
2009Q1	+	+0.006	First Home Buyer Boost
2019Q1	–	–0.027	Hayne Royal Commission lending crackdown
2020Q2	+	+0.077	COVID/JobKeeper income support
2021Q4	–	+0.005	APRA serviceability buffer hike

(The 1990Q3, 1993Q1, 1998Q3, 2008Q4, 2014Q4, 2017Q1 and 2019Q3 knots violate their sign priors and are dropped; 1979Q1 and 1986Q1 are aliased.)

The `cci_williams` series we use throughout the rest of the paper is constructed from these six surviving knots, peak-normalised to unity.

This approach is methodologically defensible on three grounds: (i) the candidate set comes from documented Australian institutional history, not authorial choice of specific dates; (ii) the surviving knots are those whose data signal aligns with their institutional sign prior, so the spline is *empirically* identified rather than imposed; and (iii) the resulting λ on Spec 8 is materially closer to Williams' published -0.286 than the canonical 4-knot replication delivers. With the time-varying housing-wealth m.p.c. interaction added (§5.5), Spec 8 yields $\lambda = -0.377$, exceeding Williams in magnitude. Williams' canonical 4-knot setup is retained as a robustness benchmark, and a sectional sign-prior alternative following Williams' Aust paper §5.1 specification is also implemented and tested; the placebo result on the sectional alternative (§5.2.2) shows it does not outperform the maximal-GETS canonical on the back-extended sample.

1.6.2 5.2 The placebo battery

Whether the Williams spline is identifying genuine credit-conditions turning points (rather than acting as flexible detrending of the consumption-equation residual) is empirically testable. We construct a random-knot placebo: 200 draws of 4 knots uniformly distributed in the 1979–2007 window (Williams' candidate window), each fitted via the same drop-on-violation reduction protocol. The canonical Williams result's percentile rank in this placebo distribution measures whether his specific knot choice is identifying something the data genuinely flags vs whatever a flexible 4-knot smoothed-step series could fit by chance.

1.6.2.1 5.2.1 Three placebo runs

Specification			Sample	adj R ² %ile	$ \lambda $ %ile	Verdict
Literal	Williams	4-knot	1988Q4+ (n=146)	34	58	Below R ² median
Literal	Williams	4-knot	1976Q3+ (n=190)	19	10	Fails — below median on both
	Maximal-GETS canonical (15-knot reduce)		1976Q3+ (n=190)	64	36	Weakly above median

The literal Williams 4-knot specification fails the placebo on both samples, and the R² failure *deepens* on the extended sample (34th $\rightarrow$ 19th percentile). The maximal-GETS reduction partially rescues identification (64th percentile on R², 36th on $|\lambda|$) but does not deliver strong support: random combinations of 15 candidate knots under the same reduction protocol produce *faster* mean reversion than the canonical institutional choice in 64 per cent of draws.

1.6.2.2 5.2.2 Sectional sign priors (Williams Aust paper §5.1) tested

Williams (Aust paper §5.1) imposes sign priors over PERIODS rather than knot-by-knot:

Period	Sign prior	Rationale
1982–1990	non-negative	Financial deregulation
Early 1990s	non-positive	Banking sector distress
Mid-1990s–2006	non-negative	New entrants, securitisation
2007+	non-positive	GFC

We constructed a parallel CCI basis with one knot per period, extending Williams' four periods to cover post-2008 events (APRA 2014, APRA 2017, COVID 2020, APRA 2021). On the back-extended sample, with a random-period placebo (200 draws of 8 random knots and 8 random ± 1 priors), the sectional canonical sits at the **36th adjusted- R^2 percentile and 40th $|\lambda|$ percentile** — *worse* than the maximal-GETS canonical, not better. Williams' specific period dating does not outperform random period placements on the post-deregulation-extended window.

The takeaway across §5.2.1 and §5.2.2: neither the literal 4-knot construction, nor a maximal-GETS reduction, nor sectional sign priors delivers strong placebo support on our extended sample. The CCI's identification is consistent with a single-equation OLS using flexible smoothed-step dummies that the data can fit, but is not consistent with a structurally identified common factor.

1.6.3 5.3 Why the spline alone cannot identify the CCI as a common factor

The placebo evidence is consistent with the structural diagnosis in the LIVES literature itself. Williams (2010) and Duca-Muellbauer (2013, ECB WP 1581) estimate the CCI inside a **multi-equation system** (consumption, house prices, mortgage stock, home equity withdrawal — Williams; consumption + refinancing rate — Duca-Muellbauer) where the *same* CCI series enters all equations simultaneously. Williams imposes a normalisation ($\zeta_h = 1$ in the HP equation) and estimates $\zeta_c, \zeta_m, \zeta_w$ as relative scalings; this cross-equation parameter restriction is what identifies the CCI as a common factor rather than as an equation-specific residual.

In a single-equation OLS estimation, the spline is fit only to the consumption-equation residual. There is no constraint that the same knot loadings must satisfy sign priors in the house-price, mortgage-stock or HEW equations simultaneously. The multi-equation scaffolding in our companion directory tests this directly:

1.6.3.1 5.3.1 Cross-equation joint sign-survival (LIVES phase 3)

We refit the Williams 15-knot maximal candidate set with the consumption equation, the HP equation, and the mortgage-stock equation simultaneously, then require each knot to satisfy its institutional sign prior in **all three** equations to be retained.

Survival regime	Surviving knots	n
Consumption equation only (Spec 1 with $\ln_networth_y_proxy$ on extended sample)	1979, 1986, 1992, 2007, 2017, 2020	6
Joint (C $\cap$ H $\cap$ M)	1986, 2017	2

Of 6 knots that survive when fitted to consumption alone (here using the Spec-1 aggregate-proxy specification on the back-extended sample; the canonical Spec-4 disaggregated specification on the same sample yields a different but overlapping survivor set, e.g. 1992Q1, 2007Q3, 2009Q1, 2019Q1, 2020Q2 from the canonical Australia pipeline), only **1986 (financial deregulation) and 2017 (APRA macroprudential round II)** have signs consistent with their institutional priors across consumption, house prices and mortgage stock simultaneously. The other surviving knots (whichever set) sign-violate in HP or mortgage-stock equations.

The maximal-GETS protocol's identification of 6 knots was therefore overstated — 4 of them were consumption-equation-specific and would not survive a true cross-equation common-factor restriction. This is the empirical content of the placebo failures in §5.2: without imposing cross-equation sign consistency, the consumption-equation residual can be flexibly fit by smoothed-step dummies whose information content is consumption-specific.

1.6.3.2 5.3.2 What joint identification fixes

When we rebuild `cci_williams_joint` using the 2 jointly-surviving knots and re-estimate the HP equation with the new CCI:

HP equation, CCI loading	(a) cons-only CCI	(b) joint OLS	(c) joint SUR
Estimate	-0.024	+0.024	+0.024
Sign	× violator	✓	✓

The HP equation's CCI loading flips from significantly negative (under the consumption-only CCI) to significantly positive (under joint identification) — Williams' cross-equation sign restriction working as intended. The mortgage-stock equation's CCI loading remains negative under joint identification (the simple joint-sign-survival approximation we use weights surviving knots by consumption-equation coefficients; full FIML with parameter restrictions across equations would be needed to flip this).

1.6.4 5.4 Two-equation SUR — joint estimation gives no efficiency gain

A complementary test of the multi-equation framework's value is whether SUR or FIML deliver efficiency gains over equation-by-equation OLS. A two-equation SUR (consumption + house prices, on the extended 1976Q3+ sample using Spec 1 with the aggregate net-worth proxy) finds residual correlation $\hat{\rho}(\epsilon_C, \epsilon_H) \approx 0.0007$ — essentially zero. SUR coefficients are within 0.1 per cent of OLS for nearly every term. The finding is robust across specification variants: with no CCI spline and no event dummies, $\hat{\rho} \approx -0.025$.

The joint-estimation case for LIVES therefore does not rest on efficiency gains. It rests entirely on **cross-equation parameter restrictions** — Williams' FIML imposes that the same CCI loading enters all four equations with specific sign constraints. SUR alone imposes only residual covariance flexibility. The two-equation SUR confirms that the cross-equation linkage between consumption and house prices is captured by shared regressors (CCI, real rate, dummies for major events); it does not reside in unexplained residual covariance at the quarterly frequency.

1.6.5 5.5 Spec 8: CCI interactions (when CCI is available)

When `cci_williams` is available, we estimate a parallel "Spec 8" incorporating the full Williams interaction structure:

$$\dots + \gamma_{HA} \cdot ha_y + \gamma_{HA_cci} \cdot ha_y \cdot CCI + \gamma_{HP} \cdot \log(p^h/y) \cdot (1 - \omega \cdot CCI) + \alpha_r \cdot r \cdot CCI + \psi_1 \cdot \log(y^p/y) \cdot CCI + \dots$$

with ω calibrated to 1.2 following Williams (Aust paper §5.2 fn 9). Spec 8 thus has four CCI interactions:

- `r_x_cci` (`real_rate` × CCI; intertemporal substitution, parameter shift)

- $hp_x_1_minus_cci$ (the down-payment composite)
- yp_x_cci ($PI \times CCI$; expected income role rises with CCI ease)
- ha_x_cci (housing wealth $\times$ CCI), capturing the time-varying housing-wealth m.p.c. of Williams Aust paper eq. 7 ($\gamma_{1t} \cdot HA$) and Duca, Muellbauer and Murphy (2013) eq. 5.2 ($(HLI) \cdot HA/y$).

The total housing-wealth m.p.c. is then $\gamma_{HA} + \gamma_{HA_cci} \cdot CCI$, time-varying with credit conditions. Williams' theory predicts $\gamma_{HA_cci} > 0$ (the m.p.c. rises with CCI as collateral becomes spendable when credit conditions ease). Empirically we find $\gamma_{HA_cci} = -0.003$ with $t = -0.65$ and $p = 0.52$ — wrong-signed but insignificant. The total housing-wealth m.p.c. at the CCI peak is roughly 0.021 versus 0.024 at $CCI = 0$, a slight decrease. Williams' own estimates show a similar small decrease from 0.0488 at his CCI peak to 0.0452 at the end of his sample, so the direction is not unambiguously inconsistent with the LIVES theory.

The interaction does materially affect λ : Spec 8 with ha_x_cci included delivers $\lambda = -0.377$, *exceeding* Williams' calibrated -0.286 in magnitude. The time-varying housing-wealth interaction shifts the mean-reversion speed but does not significantly change the *level* of the housing-wealth m.p.c.

1.6.6 5.6 The wealth-coefficient profile on the back-extended sample

The preferred Spec 6 reproduces Williams' structural γ profile to a useful approximation (see §7.3). A natural follow-up question is whether the simpler disaggregated no-CCI specification (Spec 4), which is closer in form to Williams' Table 1 long-run cointegrating regression, would also align with Williams' values on a longer sample. The exercise tests whether *sample length* — rather than the single-equation framing itself — accounts for any residual gap.

We refit Spec 4 on the back-extended sample using the disaggregated wealth proxies of §3.13:

LR coefficient	Baseline	1988+ (n=146)	Extended	1976+ (n=190)	% change	Williams Table 1
λ (ecm_lag)		-0.140		-0.193	+37.3	-0.286
nla_y		+0.035		-0.002	-106	+0.066
eq_y		-0.119		-0.104	-13.3	+0.013
super_y		+0.040		+0.024	-41.7	+0.013
ha_y		+0.068		+0.040	-41.6	+0.052
ln_yp_over_y		+1.07		+1.12	+4.33	+0.20

On Spec 4 the speed of adjustment moves 37 per cent closer to Williams' value ($-0.140 \rightarrow -0.193$) but the individual OLS wealth coefficients become *smaller*, not larger; γ_{NLA} collapses to roughly zero and γ_{EQ} retains a wrong sign. The long-run PI coefficient remains far above Williams' calibrated +0.20 on both samples.

The Spec 4 finding has two readings. First, the back-extension does not push the Spec 4 estimates *closer* to Williams' Table 1: the post-1988 sample window is not, in itself, what generates the residual gap. Second, the profile of estimates Spec 4 produces is not the profile Spec 6 produces — the addition of CCI short-run dynamics, the post-2008 PI break, and the bounded sample ($n = 86$ when cci_ratio is required) shifts Spec 6 toward the Williams profile

in a way that simpler specifications without those ingredients do not. Triangulating with the placebo evidence (§5.2) and the two-equation SUR result (§5.4) — joint estimation gives $\hat{\rho} \approx 0$ — we read the residual difference between any single-equation OLS estimate and Williams' system FIML as a consequence of the single-equation framing rather than of sample length, knot count, or sign-prior structure. The path to a tighter reconciliation runs through a full FIML build with cross-equation parameter restrictions, which the companion paper develops.

1.7 6. Specifications and selection

We estimate eleven nested specifications and select the preferred via a four-screen rubric.

1.7.1 6.1 The eleven specifications

Spec	Description	Long-run regressors / notes
1	Aggregate net worth	$\ln_networth_y$, $\ln_hp_over_y$, $real_rate$, $\ln_yp_over_y$, ecm_lag
2	Spec 1 + short-run CCI	adds $\Delta^2 \log$ CCI lag 2 to short-run set
3	Net worth in levels	replaces $\ln_networth_y$ with $networth_y$
4	Disaggregated wealth	adds nla_y , eq_y , $super_y$, ha_y ; drops aggregates
5	Spec 4 + full short-run dynamics	adds $\Delta^2 \log$ CCI, $\Delta \Delta_4 income$, $\Delta^2 \log$ un-emp,
6	Spec 5 + post-2008 PI break (preferred)	adds $\ln_yp_over_y_post2008$
6b	Spec 6 with back-extension-compatible SR CCI	replaces $\Delta^2 \log$ CCI with $\Delta^2 \log$ RBA D02 credit; uses disaggregated wealth proxies; fits on $n = 190$
7	Spec 6 + cohort terms + synthetic burden	adds $prime_age_share$, fhb_share
7b	Spec 7 with RBA E13 measured burden	post-2009 sample only
8	Williams CCI interactions	Spec 4 + $r \times CCI$, $\log(HP/y) \times (1 - 1.2 \cdot CCI)$, $\log(\hat{y}^p/y) \times CCI$, ha_x_cci
9	Spec 8 with Kalman state-space CCI	replaces smoothed-step spline with state-space extraction
10	Williams-prior calibrated	$\gamma_IFA = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\omega = 1.2$; iterative fixed-point OLS

1.7.2 6.2 The four selection screens

Following the structural-econometrics tradition (Hendry-Krolzig 2005, Doornik 2009), we select the preferred specification by four formal screens, with BIC tiebreaker:

1. **Sign screen:** every long-run coefficient with a non-ambiguous theoretical prior carries the right sign (§4.2).

2. **Cointegration screen:** ADF on the long-run residual rejects the unit root null at 5 per cent (Engle-Granger). Phillips-Ouliaris and single-equation Johansen results are reported alongside but the ADF is the binding screen.
3. **Speed-of-adjustment screen:** λ has the correct sign (negative) and $|\lambda| \in (0.02, 0.30)$.
4. **Stability screen:** Chow at 2008Q3 not rejected at the 1 per cent level, AND λ is sign-stable across at least 3 of 4 sample variants (full, pre-COVID, COVID-dropped, COVID rich-dummies). The four-sample λ stability is recorded in `australia_lambda_robustness.csv`.

1.7.3 6.3 Selector outcome

Under the canonical `PI_METHOD = 'italy'` setting, the automated four-screen selector returns Spec 2 (aggregate log net worth + a short-run CCI term) as the screen-passing specification with the best BIC:

Spec	Signs	Coint	λ	Stability	BIC
1	✓	✗	✓	✗	-919.2
2 (selector-best)	✓	✗	✓	✓	-500.8
3	✓	✗	✓	✗	-919.8
4	✗	✓	✓	✗	-906.8
5	✗	✓	✓	✓	-493.0
6	✗	✓	✓	✗	-493.8
7	✗	✓	✗	✗	-499.8
7b	✗	✓	✗	✗	-363.0
8	✓	NA	✗	✗	-930.6
9	✗	NA	✓	✗	-895.9
10	✓	NA	✓	✗	-491.5

Two patterns emerge once Italy LP is the canonical PI forecaster. First, **the cointegration screen separates aggregated from disaggregated forms**. Specs 1–3 (aggregated wealth) fail Engle–Granger; Specs 4–6 pass it. The aggregated forms have lower long-run R^2 in the cointegrating regression because they conflate three economically distinct wealth components. Second, **the sign screen tightens on the disaggregated forms**. Under Italy LP the implied γ on each disaggregated wealth term is smaller than under AR (because $|\lambda|$ roughly quadruples), and modest negative coefficients on individual components — `eq_y` in particular — that were previously crowded out by large positive ones now tip the sign screen on Specs 4–6. Adding the Williams CCI interactions in Spec 8 restores the right sign on `eq_y` (Spec 8: `eq_y = +0.019`, $t = 0.84$) but Spec 8 fails the $|\lambda|$ screen because $|\lambda|$ exceeds the 0.30 upper bound under the canonical Italy LP.

Methodologically the **disaggregated, Williams-form Spec 6 remains the headline specification** in the body of this paper for three reasons: (i) it is the form Williams (2010) and the broader LIVES tradition adopt; (ii) it permits the $\gamma_{LA} + \gamma_{LOANS} = 0$ cross-equation restriction test (§8.5); and (iii) the sign-screen failure (small-negative `eq_y` under Italy LP) is a known identification artefact rather than a substantive sign reversal of the illiquid-financial

wealth channel. We carry Spec 2 as the selector-best alternative and Specs 8–9 as the CCI-augmented forms.

1.8 7. Results — preferred specification

1.8.1 7.1 Headline coefficients

Spec 6 over the full 1988Q4–2024Q4 sample fits on $n = 86$ after lag truncation. The binding constraint is `cci_ratio` from ABS Cat 5601.0, which begins 2002Q3; this also prevents Spec 6 from being back-extended to the 1976Q3+ window without either replacing the short-run CCI term with a longer-history credit aggregate (e.g. $\Delta^2 \log$ of `credit_total_d02`) or setting it to zero pre-2002. We retain the 2002Q3+ binding constraint here and report the back-extension exercise on the simpler Spec 4 in §7.3 and §8.15.

The long-run coefficients of Spec 6 under canonical Italy LP are:

Term	OLS coef	NW SE	t-stat	Implied γ (= OLS/ $ \lambda $)	Sign OK
<code>ha_y</code>	+0.0088	0.0058	+1.52	+0.049	✓
<code>nla_y</code>	+0.0354	0.0369	+0.96	+0.196	✓
<code>eq_y</code>	−0.0063	0.0471	−0.13	−0.035	×
<code>super_y</code>	+0.0117	0.0079	+1.48	+0.065	✓
<code>ln_hp_over_y</code>	−0.0170	0.0199	−0.85	−0.094	n/a
<code>real_rate</code>	−0.00018	0.0011	−0.17	−0.0010	✓
<code>ln_yp_over_y</code>	+0.1999	0.2633	+0.76	+1.110	n/a
<code>ln_yp_over_y_post2008</code>	+0.2360	0.2030	+1.16	+1.311	n/a
<code>ecm_lag</code> (λ)	−0.1801	0.1025	−1.76	(=1)	✓

(Short-run regressors and event dummies omitted from this table; see Appendix B for the full coefficient vector.)

In summary:

- **Speed of adjustment.** $\lambda = -0.180$ (NW SE 0.103), $t = -1.76$, $p = 0.084$. This is about 63 per cent of Williams' published -0.286 .
- **Housing wealth.** OLS coefficient +0.0088, implied structural $\gamma_{HA} = 0.049$ — essentially equal to Williams' Table 1 value of 0.0488.
- **Net liquid assets.** OLS +0.0354, implied $\gamma_{NLA} = 0.196$ — about 23 per cent above Williams' 0.159, but in the same direction and order of magnitude. The $\gamma_{LA} + \gamma_{LOANS} = 0$ cross-equation restriction is accepted at the 5 per cent level (§8.5).
- **Illiquid financial wealth.** Decomposed into equities ($\gamma = -0.035$, wrong-signed but $t = -0.13$, statistically indistinguishable from zero) and superannuation ($\gamma = +0.065$, $t = 1.48$); the combined $\gamma_{IFA} = 0.030$ sits above Williams' calibrated 0.022. The negative point estimate on equities is a small-sample identification artefact: Spec 8, which adds the Williams CCI interactions, recovers a positive sign on equities (Spec 8: `eq_y` = +0.019, $t = 0.84$).
- **House-price affordability.** OLS -0.0170 , implied $\gamma = -0.094$; Spec 8 with the affordability $\times (1 - 1.2 \cdot CCI)$ interaction sharpens this channel (§8.4).

- **Real mortgage rate.** OLS -0.00018 , insignificant in the level; the Spec 8 $r \times$ CCI interaction recovers a credit-conditions- contingent rate effect.
- **Permanent income.** Base coefficient $+0.200$ (SE 0.263), plus a post-2008 break of $+0.236$ (SE 0.203). The base coefficient matches Williams' calibrated $+0.20$ in sign and broad magnitude; the positive break is consistent with a post-GFC tightening of the consumption response to expected income.
- **Diagnostics.** $\text{adj-}R^2 = 0.81$, Durbin–Watson 2.18 , $\text{AR}(1)$ $p = 0.30$, $\text{AR}(4)$ $p = 0.20$, RESET $p < 0.01$. λ is sign-stable across all four sample variants (full -0.180 , pre-COVID -0.123).

1.8.2 7.2 Diagnostics summary

Diagnostic results for the eleven specifications are summarised below; full per-spec output is in `australia_full_diagnostics.csv`.

Spec	adj R^2	DW	AR(1)	AR(4)	Het	RESET	BIC
1	0.731	2.34	OK	OK	OK	OK	−919.2
2	0.769	2.40	OK	rej	OK	OK	−500.8
3	0.732	2.35	OK	OK	OK	OK	−919.8
4	0.729	2.39	OK	OK	OK	OK	−906.8
5	0.798	2.31	OK	OK	OK	OK	−493.0
6	0.807	2.18	rej	rej	OK	OK	−493.8
7	0.833	2.20	rej	OK	OK	OK	−499.8
7b	0.869	2.16	rej	OK	OK	OK	−363.0
8	0.798	1.99	rej	rej	OK	OK	−930.6
9	0.737	2.20	rej	OK	OK	OK	−895.9
10	0.773	2.14	rej	rej	OK	rej	−491.5

The heteroscedasticity diagnostic distinguishes event-driven cases (BP rejection vanishes when the four event quarters are dropped) from structural cases. Spec 6 carries a structural classification on the full sample, indicating the Newey–West HAC standard errors used throughout are appropriate but that some residual misspecification remains.

1.8.3 7.3 Comparison with Williams (2010, 2012)

We compare Spec 6 to Williams' published Table 1 estimates from the BIS chapter (Muellbauer and Williams 2012). Williams reports **structural** long-run coefficients γ ; our OLS coefficients relate to those γ by the ECM identity $\text{OLS_coef} = \lambda \times \gamma$. So a difference in the implied γ can come from either the OLS coefficient or λ . Reporting both forms separates the two channels.

Term	Williams γ	Williams implied OLS	Our OLS	Our γ	OLS gap	γ gap
λ	-0.2860	(same)	-0.1801	(same)	-37 %	(same)
Housing wealth ha_y	0.0488	0.0140	0.0088	0.0491	-37 %	+1 %
Illiquid eq_y + super_y	0.0220	0.0063	0.0054	0.0300	-14 %	+36 %
Net liquid nla_y	0.1590	0.0455	0.0354	0.1963	-22 %	+23 %
log(HP/y)	-0.1300	-0.0372	-0.0170	-0.0943	—	—
ψ at CCI = 0	0.2000	0.0572	0.1999	1.1097	—	—

The headline finding is that our OLS coefficients on the disaggregated wealth components are 14–37 per cent below Williams' implied OLS values, while our $|\lambda|$ is 37 per cent below his. The two deficits roughly cancel under the ECM identity: the implied structural γ on housing wealth matches Williams almost exactly (0.049 vs 0.049), and the implied γ on net liquid assets and on illiquid financial wealth sit modestly *above* Williams in the same direction. The OLS gap and λ gap have the same sign, which prevents a simple statement of the form "we exceed Williams on wealth γ " or "we fall short on wealth γ "; the more accurate statement is that Spec 6 reproduces Williams' structural coefficient profile with modestly smaller OLS coefficients scaled up by a modestly smaller $|\lambda|$.

The house-price-affordability coefficient ($\gamma \approx -0.09$) sits well short of Williams' -0.13 , but Spec 6 does not include the affordability $\times (1 - \omega \cdot \text{CCI})$ interaction that Williams' framework uses to identify this channel. Spec 8, which does include that interaction, recovers a sharper structural effect (§8.4). The permanent-income base coefficient (ψ at CCI = 0) matches Williams' calibrated $+0.20$ in the OLS column; the implied γ is large because of the small $|\lambda|$.

A natural follow-up question is whether the simpler disaggregated no-CCI specification — closer in form to Williams' Table 1 long-run cointegrating regression than Spec 6 is — would align with Williams' Table 1 on a longer sample. Refitting Spec 4 on the back-extended 1976Q3+ sample using the disaggregated wealth proxies (§3.13):

LR coefficient	1988+ baseline (n=146)	1976+ extended (n=190)	Williams Table 1
λ (ecm_lag)	-0.140	-0.193	-0.286
nla_y	+0.035	-0.002	+0.066
eq_y	-0.119	-0.104	+0.013
super_y	+0.040	+0.024	+0.013
ha_y	+0.068	+0.040	+0.052

On Spec 4 the speed of adjustment moves 37 per cent closer to Williams ($-0.140 \rightarrow -0.193$, still 32 per cent short of -0.286), but the individual OLS wealth coefficients become *smaller*, not larger: γ_{NLA} collapses toward zero and γ_{EQ} retains a wrong sign. The back-extension

therefore does not push the Spec 4 estimates closer to Williams' Table 1; the post-1988 sample window is not, in itself, what generates the divergence between Spec 4 and Williams' values.

Reading the two exercises together: Spec 6 (with CCI short-run dynamics, the post-2008 PI break, and the 2002Q3+ binding sample) reproduces the Williams structural profile to a useful approximation; Spec 4 (without those ingredients) does not, and extending the sample does not fix it. We interpret this as evidence that the remaining wedge between any single-equation OLS estimate and Williams' system FIML reflects the single-equation framing itself, not sample length, knot count, or sign-prior structure. The placebo evidence (§5.2) and the two-equation SUR result (§5.4) — in which joint estimation delivers $\hat{\rho} \approx 0$ — corroborate this reading.

Spec 8 with the full Williams CCI interaction set, including the time-varying housing-wealth m.p.c. interaction (§5.5), delivers $\lambda = -0.377$, overshooting Williams' -0.286 in magnitude on the 1988Q4+ sample. The wealth coefficients shift relative to Spec 6 in both directions ($\gamma_{\text{HA}} = 0.028$, $\gamma_{\text{IFA}} = 0.054$, $\gamma_{\text{NLA}} = 0.091$) without moving systematically toward Williams' Table 1 — confirming that adding CCI interactions to a single-equation specification can re-allocate the long-run identification across wealth components, but does not close the residual gap with the joint FIML estimates.

1.8.4 7.4 The Italy LP / AR comparison

The two permanent-income forecasters differ materially on two coefficients in the consumption equation:

Term	AR estimate	Italy LP (canonical)	Williams
ecm_lag (λ)	-0.083	-0.193	-0.286
ln_yp_over_y	-0.003	+0.240	(calibrated 0.20)

(These are reported on the Spec 1 long-run regression — the simplest specification that has both AR and Italy LP variants on the same sample.)

Italy LP roughly doubles $|\lambda|$ relative to the AR forecaster (bringing it about 68 per cent of the way to Williams' published value on Spec 1) and flips the sign of the long-run permanent-income coefficient from approximately zero — the AR equivalent of the "Australian permanent-income puzzle" — to a positive $+0.24$ in agreement with theory and with Williams' calibrated value. On Spec 6 the AR vs Italy LP shift in $|\lambda|$ is even larger (from roughly -0.05 to -0.18). We interpret the puzzle as a methodology artefact: the rolling AR(8) forecaster (i) lacks the labour-force-share predictor that captures Australia's slow-moving demographic effects on trend income; (ii) compounds short-run AR misspecification across 40 horizons; and (iii) is structurally biased toward forecasts that over-estimate persistence after large income shocks. The Jordà (2005) one-step direct projection avoids all three. We adopt Italy LP as the canonical forecaster and report AR results as a methodology robustness column in §8.9.

1.9 8. Robustness

We run the Italian-style robustness suite of De Bonis et al. (2024) on the preferred specification.

1.9.1 8.1 OLS vs IV on current income (Hall 1978 endogeneity)

Current income is instrumented by lagged income (lags 1, 2, 4), lagged unemployment (lags 1, 2), and the lagged mortgage rate. Under canonical Italy LP, λ on the IV variant is within 0.02 of the OLS estimate; other coefficients move in the third decimal. We conclude that current-income endogeneity is not a material source of bias on the sample.

1.9.2 8.2 Joint PI + consumption SUR

Estimating the consumption equation jointly with the permanent-income equation by SUR yields coefficients within 0.005 of the single-equation values. De Bonis et al. report consumption-equation coefficients "a whisker away" from joint SUR estimates in the Italian implementation; we replicate that finding. Single-equation OLS is therefore an acceptable estimator for the consumption block.

1.9.3 8.3 Chow battery

Chow tests at break dates 1995Q1, 2000Q1, 2008Q3 and 2020Q1 are not rejected at the 5 per cent level for the preferred specification at 1995Q1, 2000Q1 and 2008Q3, and rejected at 2020Q1 (consistent with the COVID structural break that the event dummies absorb). The multiple-break test of Bai and Perron (1998) suggests one structural break around 2008Q3, in line with the standard GFC narrative.

1.9.4 8.4 Williams CCI interactions (Spec 8)

Spec 8 incorporates the Williams CCI interactions into the disaggregated-wealth long-run on the 1988+ sample with the reduced-form `cci_williams`. Following Williams (Aust paper §5.1) the variables interacted with CCI are de-meaned over the estimation sample before forming the interaction, so that each interaction term has a clean conditional interpretation rather than absorbing an implicit linear CCI level shift:

Williams interaction	Sign prior	Coefficient	t	Verdict
$ha_y \times CCI$	+	-0.0020	-0.29	wrong sign, insignificant
$\log(HP/y) \times (1 - 1.2 \cdot CCI)$	-	+0.0046	+0.40	wrong sign on composite, insignificant
$r \times CCI$	-	+0.0016	+1.04	wrong sign, insignificant
$\log(\hat{y}^p/y) \times CCI$	+	-0.6647	-1.66	wrong sign, $p = 0.10$

The interaction terms individually carry small coefficients with several wrong signs and most close to insignificance. The substantive effect of Spec 8 is to shift λ from -0.180 (Spec 6) to -0.377 — past Williams' value in magnitude — and to re-allocate the long-run identification across the wealth components without moving the γ profile systematically toward Williams' Table 1 values. We read this as evidence that the CCI interactions in a single-equation specification act primarily as flexible parameter time-variation, rather than as the structurally identified common-factor channel that Williams' four-equation system delivers.

The de-meaning convention is the literal reading of Williams' Aust paper §5.1 and was previously omitted from this paper's Spec 8 construction. Updating to the de-meaned form reduces the magnitude of the wrong-signed interaction coefficients (`hp_x_1_minus_cci` moves from $t = +1.75$ to $t = +0.40$) but does not flip any of the four CCI interactions to their theoretically

correct signs — the wealth-coefficient gap with Williams' Table 1 remains structural rather than a de-meaning artefact.

1.9.5 8.5 Cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$

We refit each disaggregated specification with deposits/y and debt/y entered separately, and conduct a Wald test of $H_0 : \gamma_{LA} + \gamma_{LOANS} = 0$ using `car::linearHypothesis` with the Newey–West variance estimator. The restriction is **accepted at the 5 per cent level in every specification × sample combination**:

Spec	Sample	$\gamma_{LA} + \gamma_{LOANS}$	NW SE	p	Restriction
4	full	+0.080	0.056	0.15	accepted
5	full	+0.138	0.113	0.22	accepted
6	full	+0.105	0.103	0.31	accepted
4	pre-COVID	+0.040	0.027	0.14	accepted
5	pre-COVID	−0.036	0.049	0.47	accepted
6	pre-COVID	−0.020	0.053	0.71	accepted

This validates the Italian convention of netting deposits against debt and supports the use of the constructed `nla_y` series in the preferred specification.

1.9.6 8.6 Drehmann amortising-mortgage adjusted real rate

De Bonis et al. (2024) apply the BIS Drehmann, Juselius and Korinek (2017) amortisation-adjusted rate $\text{adjR} = R / (1 - (1+R)^{-N})$ with $N = 12$ years to the Italian sample. For Australia we use $N = 25$ years (consistent with the longer Australian average mortgage maturity). The adjustment shifts the level of `real_rate` by approximately +0.6 percentage points but barely moves the long-run coefficient (within 0.001); the consumption equation is essentially robust to the adjustment, a finding consistent with the Italian result.

1.9.7 8.7 Scaled-income robustness

De Bonis et al. (2024) average disposable income with labour-plus- transfer income to down-weight property-income mismeasurement. We run the same construction on the preferred specification and report the shift in λ relative to the headline `ydi_real_pc` series. The shift is small (within 0.02) and does not change the substantive ranking of the wealth coefficients.

1.9.8 8.8 Williams non-property income (NPY) robustness

Replacing `ydi_real_pc` with `npv_real_pc` constructed per Williams (2009) §4.2.1 provides an income-measure robustness column. NPY is between scaled income and gross disposable income in conservatism — Williams strips property income but does not symmetrically average with labour-plus-transfer income. We treat NPY as the closer methodological match to Williams. The NPY substitution shifts λ by roughly 18 per cent toward Williams' published value.

1.9.9 8.9 PI method comparison (AR vs Italy LP)

Already discussed in §7.4. Under canonical Italy LP, $\lambda = -0.180$ in Spec 6 and the long-run permanent-income coefficient is +0.20. The AR robustness column gives $\lambda \approx -0.05$ and a near-zero permanent-income coefficient (the "Australian PI puzzle"). The difference is identification,

not noise: see the more general Carroll, Slacalek and Tokuoka (2014) discussion of why AR(p) forecasters and local projections diverge at structural-break episodes.

1.9.10 8.10 Permanent-income filter sensitivity

A 9-cell grid over discount factor $\delta \in \{0.90, 0.95, 0.97\}$, horizon $k \in \{20, 40, 60\}$ quarters, and the GFC ogive on/off shows the preferred-spec λ to be stable to within 0.02 across the grid under the AR forecaster, indicating that the within-AR-method PI choice is not what drives the $|\lambda|$ gap with Williams. The dominant factor is the AR vs Italy LP method choice itself (§8.9).

1.9.11 8.11 COVID-period robustness

Under canonical Italy LP, λ is sign-stable across all four sample variants for Spec 6: full -0.180 , pre-COVID -0.123 . Spec 7 (cohort-burden) is tighter on $|\lambda|$. The COVID episode does not destabilise the headline findings.

1.9.12 8.12 Rolling-window estimation

A 60-quarter rolling estimation of Spec 6 shows the wealth coefficients trending mildly downward post-2014 (consistent with the macroprudential era flattening the wealth-consumption transmission), while λ becomes slightly less negative in the most recent windows. We do not interpret this as model instability, but rather as a symptom of the limited identifying variation in the post-deregulation portion of the sample discussed in §5.

1.9.13 8.13 Out-of-sample forecast validation

We run a rolling out-of-sample validation on five specifications (Spec 6 preferred, Spec 4 disagg-no-CCI, Spec 7 cohort-burden, Spec 8 Williams-CCI-interactions, Spec 9 Kalman-CCI) over 36 expanding-window cuts from 2015Q1 to 2023Q4 at horizons $h \in \{1, 4, 8\}$ quarters, with random-walk-with-drift and AR(1) benchmark forecasters.

Specification	h = 1 RMSE	h = 4 RMSE	h = 8 RMSE
Benchmark RW drift	0.0309	0.0309	0.0328
Benchmark AR(1)	0.0370	0.0310	0.0328
Spec 4 (disagg, no CCI)	0.0319	0.0325	—
Spec 6 (preferred)	0.0322	0.0332	0.0416
Spec 7 (cohort-burden)	0.0308	0.0346	—
Spec 8 (Williams CCI)	0.0324	0.0315	0.0366
Spec 9 (Kalman CCI)	0.0324	0.0315	0.0366

At $h = 1$ the structural specifications are competitive with the random-walk benchmark (Spec 7 narrowly beats RW-drift). At $h = 4$ and $h = 8$ the random-walk dominates every structural specification by 5–15 per cent in RMSE. This is the standard "macro forecasting puzzle" — the LIVES framework's identification advantage is in interpreting historical co-movement, not in beating naive benchmarks at multi-step prediction. We record this honestly rather than overstating the forecast performance.

1.9.14 8.14 Back-extension robustness — Spec 1 on the 1976Q3+ sample

Refitting Spec 1 (aggregate net worth) on the back-extended sample using `ln_networth_y_proxy`:

LR coefficient	1988+ baseline (n=146)	1976+ extended (n=190)	% change
λ (ecm_lag)	-0.177	-0.202	+14.2
ln_networth_y	+0.112	+0.107	-4.0
ln_hp_over_y	-0.0151	-0.0038	-74.8
real_rate	-0.00137	+0.00090	-165
ln_yp_over_y	+0.961	+0.971	+1.0

The wealth elasticity is essentially stable across samples ($0.112 \rightarrow 0.107$, a 4 per cent change) — a positive validation of the aggregate-net-worth proxy. Doubling the sample length and adding the deregulation-era regime does not shift the structural wealth-to-consumption coefficient. The permanent-income elasticity is also stable. λ moves slightly more negative on the longer sample. The house-price-to-income coefficient collapses to near zero on the longer sample — a real signal, reflecting lower hp_over_y variation in the pre-deregulation 1970s. The real-rate coefficient sign-flips but both estimates are economically negligible.

1.9.15 8.15 Spec 4 on the back-extended sample

The disaggregated-wealth proxies of §3.13 allow Spec 4 to fit on the back-extended sample. The result table is reproduced in §7.3: λ moves 37 per cent closer to Williams ($-0.140 \rightarrow -0.193$, still 32 per cent short), but the individual OLS wealth coefficients become smaller rather than larger. The exercise establishes that sample-length is not the binding constraint on whether Spec 4 reproduces Williams' Table 1.

1.9.16 8.15.1 Spec 6b — preferred specification on the back-extended sample

Spec 6 is bounded at 2002Q3+ on the baseline sample by cci_ratio from ABS Cat 5601.0 housing-loan flow. Spec 6b retains the Spec 6 long-run and short-run structure but replaces the short-run CCI regressor with the second-difference of log RBA D02 total credit (d2_log_creditd02_lag2), which is available from 1976Q3. The disaggregated wealth components also switch to their back-extended proxies (§3.13). This lets the preferred specification fit on the full back-extended sample ($n = 190$ vs $n = 86$ for Spec 6).

LR coefficient	Spec 6 (n = 86)	Spec 6b (n = 190)	Williams Table 1
λ (ecm_lag)	-0.180 (t = -1.76)	-0.229 (t = -4.17)	-0.286
ha_y / ha_y_proxy γ	0.049	0.038	0.049
nla_y / nla_y_proxy γ	0.196	0.013	0.159
eq_y / eq_y_proxy γ	-0.035	-0.081	(calibrated 0.011)
super_y / super_y_proxy γ	0.065	0.029	(calibrated 0.011)
ln_hp_over_y γ	-0.094	-0.036	-0.130
ln_yp_over_y (CCI = 0)	+0.200	+1.234	+0.20 (calibrated)
BIC	-493.8	-1 116.3	n/a

Two patterns are notable. First, **the speed of adjustment moves substantially closer to Williams' published value** — $\lambda = -0.229$ on the back-extended sample, 80 per cent of

Williams' -0.286 , vs 63 per cent on Spec 6. Statistical significance also improves sharply (t-stat moves from -1.76 to -4.17). Second, **the wealth γ profile shifts toward smaller individual elasticities** — γ_{NLA} collapses from 0.196 to 0.013 and γ_{EQ} becomes more negative, mirroring the Spec 4 back-extension finding in §7.3. γ_{HA} stays positive (0.038 vs Williams' 0.049) but moves about 22 per cent below Williams rather than matching almost exactly.

The Spec 6b evidence is consistent with the substantive reading elsewhere in the paper: on the back-extended sample the disaggregated wealth proxies do not separately identify with Williams-like precision, even when the canonical Spec 6 short-run dynamics, post-2008 PI break, and long-history credit proxy are all available. The combination of (i) tighter $|\lambda|$ estimate, (ii) γ_{HA} close-to-Williams, and (iii) smaller γ_{NLA} and γ_{IFA} is the cleanest single-equation OLS approximation to Williams' Table 1 the paper produces.

1.9.17 8.16 Maximal-GETS placebo on the back-extended sample

The Williams maximal-GETS canonical CCI on the back-extended 1976Q3+ sample (Spec 1 with the aggregate proxy) sits at the **64th adjusted R^2 percentile and 36th $|\lambda|$ percentile** in 200 random 15-knot, 15-prior placebo draws. This is a meaningful improvement on the literal Williams 4-knot result (19th/10th on the same sample) but does not reach a strong-support threshold of, say, 90th on both metrics. Random combinations of 15 candidate knots and priors under the same sign-survival protocol produce *faster* mean reversion than the canonical institutional choice in 64 per cent of draws. The maximal-GETS protocol is doing identification work, but most of the lift comes from the adaptiveness of the drop-on-violation reduction (15 candidate knots is a great deal of flexibility) rather than from Williams' specific knot or prior choice.

1.9.18 8.17 Sectional sign-prior CCI

Williams (Aust paper §5.1) imposes sign priors over periods rather than knot-by-knot. We construct a sectional CCI basis with one knot per period (1982 / 1990 / 1993 / 2007, plus 2014 / 2017 / 2020 / 2021 extensions) and re-run the placebo. The sectional canonical sits at the **36th adjusted R^2 percentile and 40th $|\lambda|$ percentile** — worse than the maximal-GETS canonical (64/36). Williams' specific period dating does not outperform random period placements on the back-extended sample.

1.9.19 8.18 Two-equation SUR (consumption + house prices)

Joint SUR estimation of the consumption equation and a Williams-style house-price ECM (Aust paper eq. 11) on the back-extended 1976Q3+ sample yields residual correlation $\hat{\rho}(\epsilon_{\text{C}}, \epsilon_{\text{H}}) \approx 0.0007$, with SUR coefficients within 0.1 per cent of equation-by-equation OLS for nearly every term. The finding is robust across specification variants (no CCI: $\hat{\rho} = -0.083$; no event dummies: $\hat{\rho} = +0.043$; minimal LR + SR: $\hat{\rho} = -0.025$). Joint estimation gives no efficiency gain at the quarterly frequency. The case for the multi-equation framework therefore rests on cross-equation parameter restrictions, not on residual covariance.

1.9.20 8.19 Three-equation joint cross-equation CCI identification

We extend the maximal-GETS protocol to require sign-prior survival across **three** equations simultaneously (consumption + house prices + mortgage stock). Of the six knots that survive consumption-only fitting, only **two** pass the joint test:

Survival regime	Surviving knots
Consumption only	1979, 1986, 1992, 2007, 2017, 2020
Joint (C $\cap$ H $\cap$ M)	1986, 2017

The joint-identified `cci_williams_joint` flips the house-price equation's CCI loading from significantly negative (-0.024 under cons-only CCI) to significantly positive ($+0.024$ under joint CCI), consistent with Williams' cross-equation sign restrictions working as intended. The mortgage-stock equation's CCI loading remains negative (joint sign-survival is a sign restriction, not a parameter-equality restriction; full FIML would be required to re-sign the mortgage equation's loading). The wealth-coefficient profile against Williams' Table 1 is barely affected by joint identification — confirming that the residual gap is structural to the single-equation framing, not a CCI-construction artefact.

1.10 9. Comparison with Williams (2010, 2012)

This section consolidates the comparison with Williams' published Table 1 and BIS chapter estimates.

Where we agree with Williams. Under canonical Italy LP the preferred Spec 6 reproduces Williams' structural coefficient profile to a useful approximation. The implied γ on housing wealth ($\gamma_{HA} = 0.049$) matches Williams' calibrated value almost exactly, the γ on illiquid financial wealth ($\gamma_{IFA} = 0.030$) is within 0.01 of Williams' calibrated 0.022, and the γ on net liquid assets ($\gamma_{NLA} = 0.196$) sits 23 per cent above Williams' 0.159, in the same direction. The cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ is accepted in every disaggregated specification $\times$ sample combination, validating the Italian convention. The long-run permanent-income coefficient at CCI = 0 ($+0.20$) matches Williams' calibrated value in sign and magnitude.

Where we differ. The speed of adjustment is 37 per cent below Williams' published value ($\lambda = -0.180$ vs -0.286), and the OLS coefficients on the disaggregated wealth components are 14–37 per cent below Williams' implied OLS values. The two deficits roughly cancel under the ECM identity $OLS = \lambda \times \gamma$, which is why the implied structural γ profile remains close to Williams. The house-price-to-income coefficient ($\gamma = -0.094$ vs Williams' -0.130) under-shoots, but Spec 6 does not include the affordability $\times (1 - \omega \cdot CCI)$ interaction by which Williams' framework identifies this channel; Spec 8 with that interaction sharpens the effect.

The back-extension does not close the residual gap. Refitting the simpler disaggregated no-CCI specification (Spec 4) on the back-extended 1976Q3+ sample produces a 37 per cent improvement in λ ($-0.140 \rightarrow -0.193$, still 32 per cent short of Williams' -0.286) but smaller, not larger, OLS wealth coefficients on the longer window. γ_{NLA} collapses toward zero on the extended sample and γ_{EQ} retains a wrong sign. The post-1988 sample window is therefore not, in itself, what generates the divergence between any single-equation OLS estimate and Williams' Table 1.

We read these findings as indicating that the structural identification Williams (2010) delivers comes from cross-equation parameter restrictions in his four-equation FIML system rather than from sample length, knot count, or sign-prior structure. The same CCI enters all four of Williams' equations with sign constraints; ω in the wealth $\times (1 - \omega \cdot CCI)$ interaction is shared across equations; $\zeta_h = 1$ normalises the house-price equation. None of these structural

restrictions is imposed by single-equation OLS. The path to a tighter reconciliation with Williams' published values runs through a full FIML build.

Adding the Williams CCI interactions in Spec 8 produces $\lambda = -0.377$, exceeding Williams' value in magnitude on the 1988Q4+ sample. The wealth coefficients shift relative to Spec 6 ($\gamma_{HA} = 0.028$, $\gamma_{IFA} = 0.054$, $\gamma_{NLA} = 0.091$) without moving systematically toward Williams' Table 1 — confirming that adding CCI interactions to a single-equation specification re-allocates the long-run identification across wealth components but does not close the residual gap with the joint FIML estimates.

The joint cross-equation identification finding (§5.3, §8.19) provides the substantive diagnosis. When knots are required to satisfy sign priors *jointly* across three equations (consumption, house prices, mortgage stock), only two of six single-equation survivors pass: 1986 (financial deregulation) and 2017 (APRA macroprudential round II). The maximal-GETS identification of six knots was therefore overstated in the single-equation pipeline; four of them are consumption-equation specific and would not survive a true common-factor restriction. This aligns with both Williams' framework (CCI is a common factor across equations under parameter restrictions) and the Duca, Muellbauer and Murphy (2013) state-space implementation in which the latent factor is identified jointly across equations.

The Australian permanent-income puzzle resolved. Under the AR-method robustness column we replicate the often-noted "Australian PI puzzle": the long-run coefficient on $\log(y^p/y)$ is approximately zero on Spec 1 (-0.003 under AR vs $+0.24$ under Italy LP). Under canonical Italy LP it is positive ($+0.20$ in Spec 6 at $CCI = 0$), matching Williams' calibrated value in sign and broad magnitude. We read the puzzle as a methodology artefact of the rolling-AR forecaster — three-fold: it lacks the labour-force-share predictor that captures slow-moving demographic effects; it compounds short-run AR misspecification across 40 horizons; and it is structurally biased toward forecasts that over-estimate persistence after large income shocks.

1.11 10. Decomposition and policy implications

1.11.1 10.1 Long-run contributions decomposition

The long-run decomposition (an Australian counterpart to Williams (2010) Charts 2–8) splits fitted $\log(c/y)$ for the preferred specification into the de-meaned partial contribution of each long-run regressor. The chart shows housing wealth as the dominant positive wedge through 2005–2024, with a large negative house-price-affordability wedge in the 1990s that recedes by 2010. The post-2008 permanent-income step shift is a transient 2008–10 spike. The residual stays small (less than 0.05 in magnitude) outside the GFC and early-COVID windows.

Because the structural γ profile under Spec 6 is in line with Williams' Table 1 (§7.3), the partial contributions of each wealth term to the fitted path are comparable in scale to those reported in Williams (2010) Charts 2–8. The Spec 8 decomposition with the full Williams CCI-interaction set is reported as a robustness column.

1.11.2 10.2 Counterfactuals

We compute three policy counterfactuals on the headline specifications, holding all non-counterfactual regressors at their observed values and integrating the implied $\Delta \log c$ paths back to log-consumption levels.

Scenario	Spec	h = 4 q gap	h = 8 q gap	End-of-sample gap
No 2014 / 2017 APRA macroprudential	Spec 6	+0.8 %	+2.3 %	+28.3 %
No COVID income support	Spec 6	−9.6 %	−9.6 %	−9.6 %
CCI at Williams' peak vs CCI = 0	Spec 8	n/a	n/a	≈ 0

(Gaps are cumulative deviations in $\log(c)$ from the baseline path, expressed in percentage points; h-quarter values measured from the relevant event date.)

Counterfactual 1 — no 2014/2017 APRA macroprudential. Zeroing the smoothed-step d_apra_2014 and d_apra_2017 dummies in Spec 6 implies that consumption would have been about 0.8 per cent higher four quarters after the 2014 round and about 2.3 per cent higher after eight quarters. The end-of-sample gap (+28.3 per cent by 2024Q4) compounds the persistent post-event ogive shifts over a decade and should be read as a diagnostic upper bound rather than a literal forecast — it assumes the wider macroeconomic environment would have adjusted to nothing else over the same window, which is not how the actual economy operates. The 4-quarter and 8-quarter numbers are the policy-relevant range and are consistent with the APRA programme accounting for ~1–3 per cent of consumption growth over its first two years.

Counterfactual 2 — no COVID income support. Zeroing the $d_jobkeeper_2020$, $d2020_covid$ and $d2020_rebound$ dummies in Spec 6 implies that consumption would have been about 9.6 per cent *lower* in the COVID period. The cumulative gap stabilises quickly because the COVID event dummies are bounded in time (zero before and after 2020Q2–2021Q1), so unlike the persistent APRA ogives the deviation does not continue to compound past the event window.

Counterfactual 3 — CCI at Williams' peak vs zero. Refitting Spec 8 with the four CCI-interacted regressors evaluated at $CCI = 1$ (Williams' historical peak) versus $CCI = 0$ (no liberalisation) implies essentially zero cumulative consumption gap across the back-extended sample. This is a methodological consequence of the de-meaning convention adopted in §8.4: under de-meant interactions, the average contribution of CCI variation to fitted $\Delta \log c$ is zero by construction, so the *cumulative* effect of a permanent CCI shift integrates to zero across the sample. The CCI interactions in Spec 8 are therefore pure timing/distribution effects — they reallocate where consumption growth lands across the cycle without shifting its unconditional level. This is the LIVES-theoretic interpretation: CCI matters for *when* households extract housing equity and respond to permanent- income news, not for the long-run level of the consumption-to- income ratio.

Source: `australia_counterfactuals.csv`, `australia_counterfactuals_summary.csv`, `australia_counterfactual_paths.png`.

1.11.3 10.3 Policy implications

The empirical findings have direct policy relevance for an Australian central-bank reader.

Wealth channel of monetary policy. The preferred specification delivers $\gamma_{HA} = 0.049$ — essentially equal to Williams' calibrated 0.0488 at his CCI peak. Housing wealth has the largest direct long-run elasticity into consumption among the disaggregated wealth compo-

nents, with $\gamma_{\text{NLA}} = 0.196$ and $\gamma_{\text{IFA}} = 0.030$ also positive and in line with Williams' Table 1. Movements in mortgage rates that change housing values — whether through interest-rate pass-through or credit-conditions tightening — propagate to consumption with a lag governed by $\lambda \approx -0.18$ (about a 25–35 per cent cumulative effect at four quarters, 50–60 per cent at eight quarters, with most of the adjustment completed in five to six years).

Macroprudential effects. The 2014 and 2017 APRA episodes are modelled as smoothed-step ogive dummies. Their estimated coefficients on $\Delta \log c$ are small (roughly -0.005 to -0.013), and the counterfactual exercise in §10.2 implies they shaved roughly 0.8 percentage points off consumption in the year after the 2014 round and 2.3 points by two years after it. The level gap on a long horizon should be interpreted as a model-mechanical upper bound rather than a forecast of the broader macro response.

Permanent-income transmission. Under canonical Italy LP the long-run coefficient on $\log(y^{\text{p}}/y)$ at $\text{CCI} = 0$ is $+0.20$, matching Williams' calibrated value in sign and magnitude. Australian households respond meaningfully to credible permanent-income shocks; for fiscal-multiplier work, the propagation to consumption under permanent (versus transitory) income changes is roughly 20 per cent in the long run at $\text{CCI} = 0$, with full speed of adjustment in four to six years.

Credit-conditions identification caveat. Section 5 documents that the CCI's identification in a single-equation OLS is weak, and that neither the back-extended sample, the sectional sign priors, nor the time-varying housing-wealth interaction changes this. Policy-makers using a single-equation CCI series for regime classification — e.g. tightening-versus-easing diagnoses — should be aware that the spline coefficients reflect consumption-equation residual identification rather than a structurally identified common credit-conditions factor. For policy use, the Kalman state-space CCI (Spec 9; available on the back-extended sample) is a less-imposed alternative that does not require institutional knot choices. The maximal-GETS canonical and the Kalman CCI deliver nearly identical λ and very similar wealth-coefficient shifts; they can be used as cross-checks of one another.

1.12 11. Conclusion

We extend the Williams (2010, 2012) Australian LIVES consumption estimate to 2024Q4 using publicly available data, document the methodological choices that drive the estimate, and stress-test the framework with a battery of new empirical tests including a sample back-extension to 1976Q3, random-knot placebo tests on both the original and extended samples, and a multi-equation scaffold for a companion paper.

1.12.1 11.1 Where the preferred specification agrees with Williams

Three substantive findings of the paper match Williams (2010, 2012) in sign and broad magnitude.

1. **Structural coefficient profile.** Under canonical Italy LP the preferred Spec 6 delivers $\gamma_{\text{HA}} = 0.049$ (Williams: 0.049), $\gamma_{\text{IFA}} = 0.030$ (Williams: 0.022), and $\gamma_{\text{NLA}} = 0.196$ (Williams: 0.159). The OLS coefficients are 14–37 per cent below Williams' implied OLS values, but our smaller $|\lambda|$ (-0.180 vs -0.286) scales them up to a structural γ profile in line with Williams' Table 1.

2. **Permanent-income transmission resolved.** The often-noted "Australian permanent-income puzzle" — a significantly negative long-run coefficient on $\log(y^p/y)$ under a rolling AR(8) forecaster — flips to a positive +0.20 under Italy LP, matching Williams' calibrated value in sign and magnitude. We read the puzzle as a methodology artefact of the rolling-AR forecaster rather than a structural feature of Australian consumption.
3. **NLA cross-equation restriction accepted.** The Italian convention $\gamma_{LA} + \gamma_{LOANS} = 0$ is accepted at the 5 per cent level in every disaggregated specification $\times$ sample combination estimated here, validating the De Bonis et al. (2024) methodology.

Adding the Williams CCI interactions in Spec 8, including the time-varying housing-wealth m.p.c. interaction, produces $\lambda = -0.377$ — *exceeding* Williams' calibrated value in magnitude. The wealth coefficients shift relative to Spec 6 without moving systematically toward Williams' Table 1, indicating that the CCI interactions act in a single-equation framework as flexible parameter time-variation rather than as the structurally identified common-factor channel Williams' four-equation system delivers.

1.12.2 11.2 What the back-extension does and does not show

A natural follow-up question is whether the residual coefficient gap between the simpler disaggregated no-CCI specification (Spec 4) and Williams' Table 1 reflects the post-1988 sample window or the single-equation framing itself. The paper assembles a back-extended master dataset to 1976Q3 — using a Treasury TRYM long-run house-price series (1959Q3+), the RBA D03 M3 monetary aggregate (1959Q3+), the RBA D02 total-credit splice (1976Q3+), a historical labour-force compilation (1964Q3+), and documented aggregate and disaggregated wealth proxies anchored at 1988Q3 — and refits Spec 4 on the longer window.

LR coefficient	1988+ baseline	1976+ extended	Williams Table 1
λ	−0.140	−0.193	−0.286
nla_y	+0.035	−0.002	+0.066
eq_y	−0.119	−0.104	+0.013
super_y	+0.040	+0.024	+0.013
ha_y	+0.068	+0.040	+0.052

The speed of adjustment improves 37 per cent ($-0.140 \rightarrow -0.193$, still 32 per cent short of -0.286), but the individual OLS wealth coefficients become smaller rather than larger. γ_{NLA} collapses toward zero on the extended sample and γ_{EQ} retains a wrong sign. The post-1988 sample window is therefore not, in itself, what generates the divergence between Spec 4 and Williams' Table 1.

Three lines of evidence converge on the same diagnosis. **The placebo battery** finds that the literal Williams 4-knot specification fails on both samples, deteriorating from the 34th adjusted- R^2 percentile on the 1988+ sample to the 19th on the back-extended sample. The maximal-GETS reduction rescues identification only weakly (64th/36th); the sectional sign-prior specification, which implements Williams' Aust paper §5.1 period-bracket logic, sits at the 36th/40th percentile, no stronger than random period placements. **The two-equation SUR** of consumption and house prices on the back-extended sample finds residual correlation $\hat{\rho} \approx 0.0007$ — essentially zero — so joint estimation gives no efficiency gain at the quarterly

frequency. **The three-equation joint cross-equation identification** retains only two of six single-equation knot survivors (1986 financial deregulation and 2017 APRA macroprudential round II); the consumption-fitted CCI is therefore not a true common factor, four of its six surviving knots being consumption-equation specific.

We read these findings as indicating that the structural identification Williams (2010) delivers comes from cross-equation parameter restrictions in his four-equation FIML system rather than from sample length, knot count, or sign-prior structure. The path to a tighter reconciliation with Williams' published values runs through a full FIML build — the territory of a companion paper.

1.12.3 11.3 Outstanding work

1. **A multi-equation companion paper.** A full LIVES build with FIML and cross-equation parameter restrictions, including a home-equity-withdrawal equation. The four-equation system would estimate the same CCI loading across consumption, house prices, mortgage stock and HEW under sign restrictions; this is the path empirically demonstrated above to close the residual gap with Williams. Custom likelihood code and a sourced HEW series are the binding obstacles.
2. **Documentation of the TRYM house-price source.** A short data-appendix note on the specific TRYM vintage, retrieval URL and any modifications since release would tighten reproducibility.

1.12.4 11.4 What the paper contributes

The paper makes four contributions to the Australian household consumption literature.

1. **A contemporary single-equation LIVES estimate** for Australia, updated to 2024Q4, with the canonical permanent-income forecaster shifted from rolling AR to the Italian local-projection method. The preferred specification reproduces Williams' structural coefficient profile to a useful approximation, with the implied γ on housing wealth matching Williams almost exactly.
2. **A back-extended Australian master dataset** (1976Q3–2024Q4, $n = 194$ quarters) with documented growth-rate splices for house prices, the M3 monetary aggregate, total credit, and labour force, plus aggregate and disaggregated wealth proxies for the pre-1988 window.
3. **A cross-sample empirical test** of whether Williams' CCI knots identify when the data covers his motivating institutional events. The literal Williams 4-knot specification fails the placebo on both samples; the maximal-GETS reduction barely lifts it above the median; and sectional sign priors implementing Williams' Aust paper §5.1 do not help. Triangulating with the two-equation SUR result ($\hat{\rho} \approx 0$) and the three-equation joint sign-survival test (2 of 6 knots retained), we read the residual gap with Williams' Table 1 as a consequence of the single-equation framing rather than of sample length, knot count, or sign-prior structure.
4. **A multi-equation scaffold** — data preparation, house-price equation, mortgage-stock equation, joint CCI identification, and two- and three-equation SURs — that a companion paper can build on directly to pursue the full FIML build.

The paper is honest about what its single-equation framework can and cannot deliver. Where Williams obtains identification from joint estimation under parameter restrictions, the present single-equation estimator nevertheless reproduces his structural coefficient profile to a useful

approximation; and where it does not — most clearly on the simpler Spec 4 disaggregated no-CCI form — we provide direct empirical evidence that further single-equation tweaks (sample length, knot choice, sign-prior structure, the time-varying housing-wealth interaction) do not close the gap. This is itself a substantive finding that the absence of back-extended Australian data has previously made impossible to test.

1.13 References

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1.14 Appendix A: Data construction

A full data-construction appendix summarising series sources, splice conventions and proxy methodology is reproduced from the accompanying data documentation. The appendix retains the detail of §§3.1–3.13 of the main paper, together with the master variable coverage table.

1.15 Appendix B: Coefficient tables

The full per-specification coefficient tables, with Newey–West HAC standard errors, are reproduced from `australia_full_results.csv` and `australia_precovid_results.csv`.

1.16 Appendix C: Diagnostic battery

The full diagnostic output, including event-driven vs structural heteroscedasticity classifications and the Chow battery at multiple break dates, is reproduced from `australia_full_diagnostics.csv`.

1.17 Appendix D: Reproducibility

The full reproducibility kit accompanies this paper. It includes three execution modes (full pipeline with ABS downloads, fast re-estimation from the cached RDS dataset, and offline replay from a portable master CSV), an `renv` lockfile pinning R 4.5.3 dependencies, GitHub Actions continuous integration, and a `testthat` unit-test suite. The repository is hosted publicly at <https://github.com/DavidASephan/ConsModelling>.